Cartographic Visualization: Visualizing Spatial Phenomena

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Slides adapted from Sarah Battersby

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Getting started with map-making

1. What kind of map should I make? (Where does my phenomena occur, and how does it change?)

2. How do I represent the data correctly?
    1. Using the right marks (at the right scale)
    2. Choosing a classification method
    3. Choosing a projection

What kind of map?

How does phenomenon change?

Where does phenomenon occur?

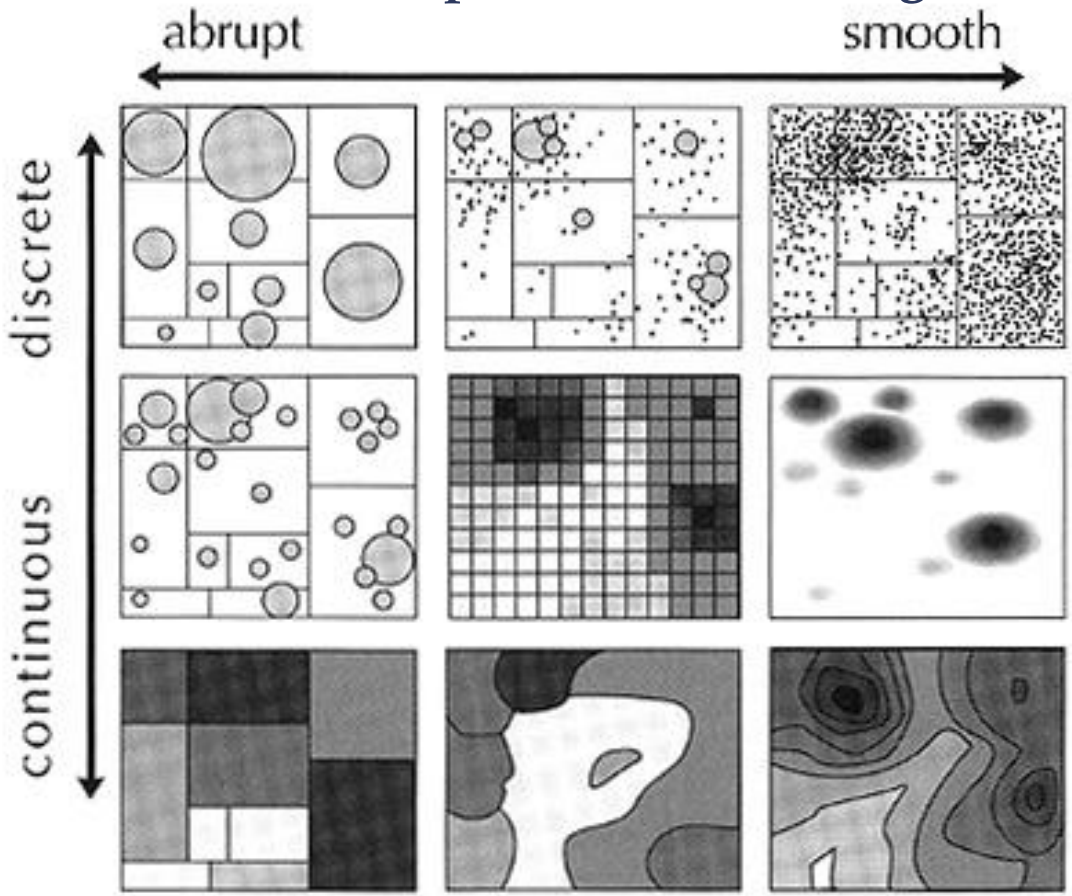
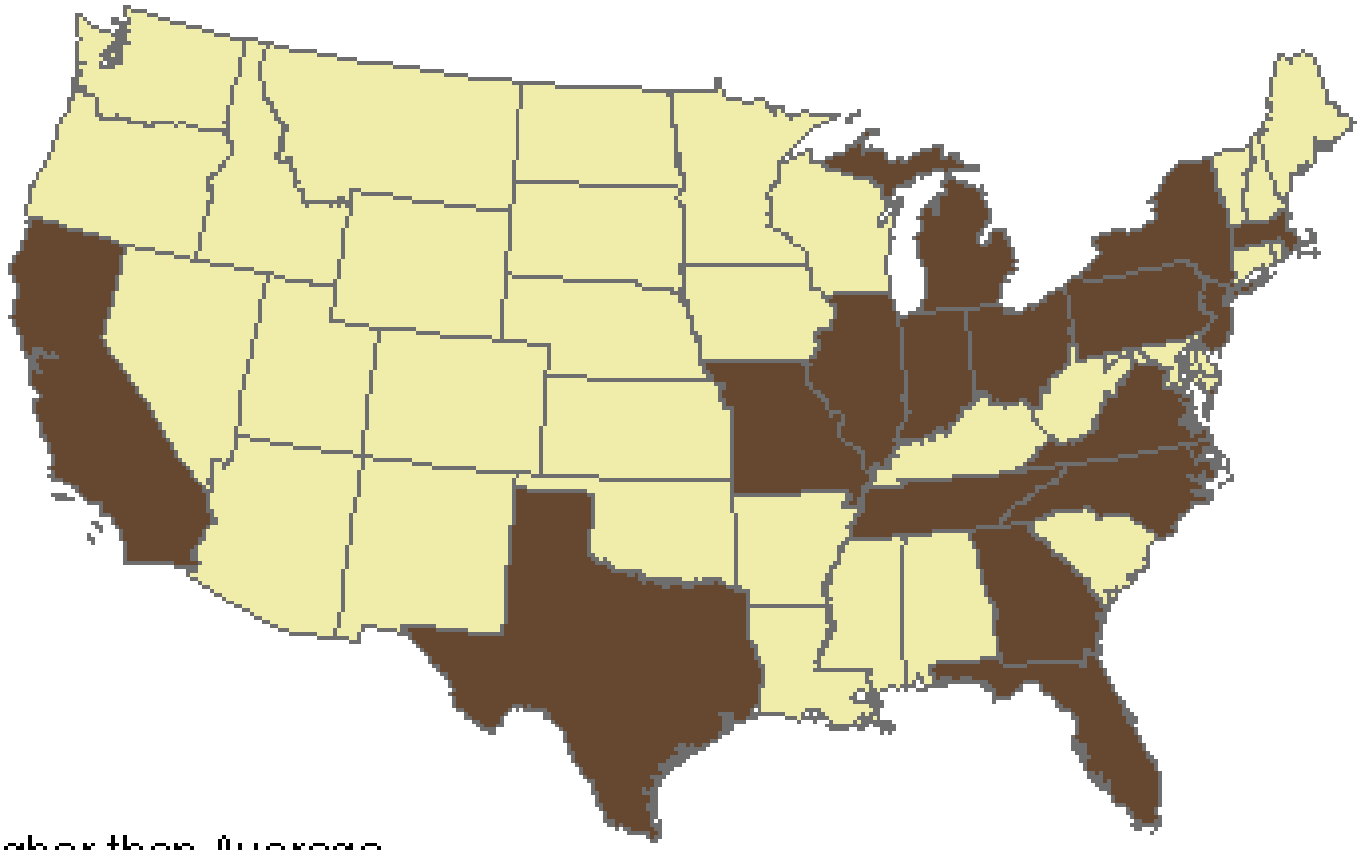


Figure 3.24 Matching map types to data models. Reproduced from MacEachren (1992) with permission.

Total Number of Melanoma-Related Deaths



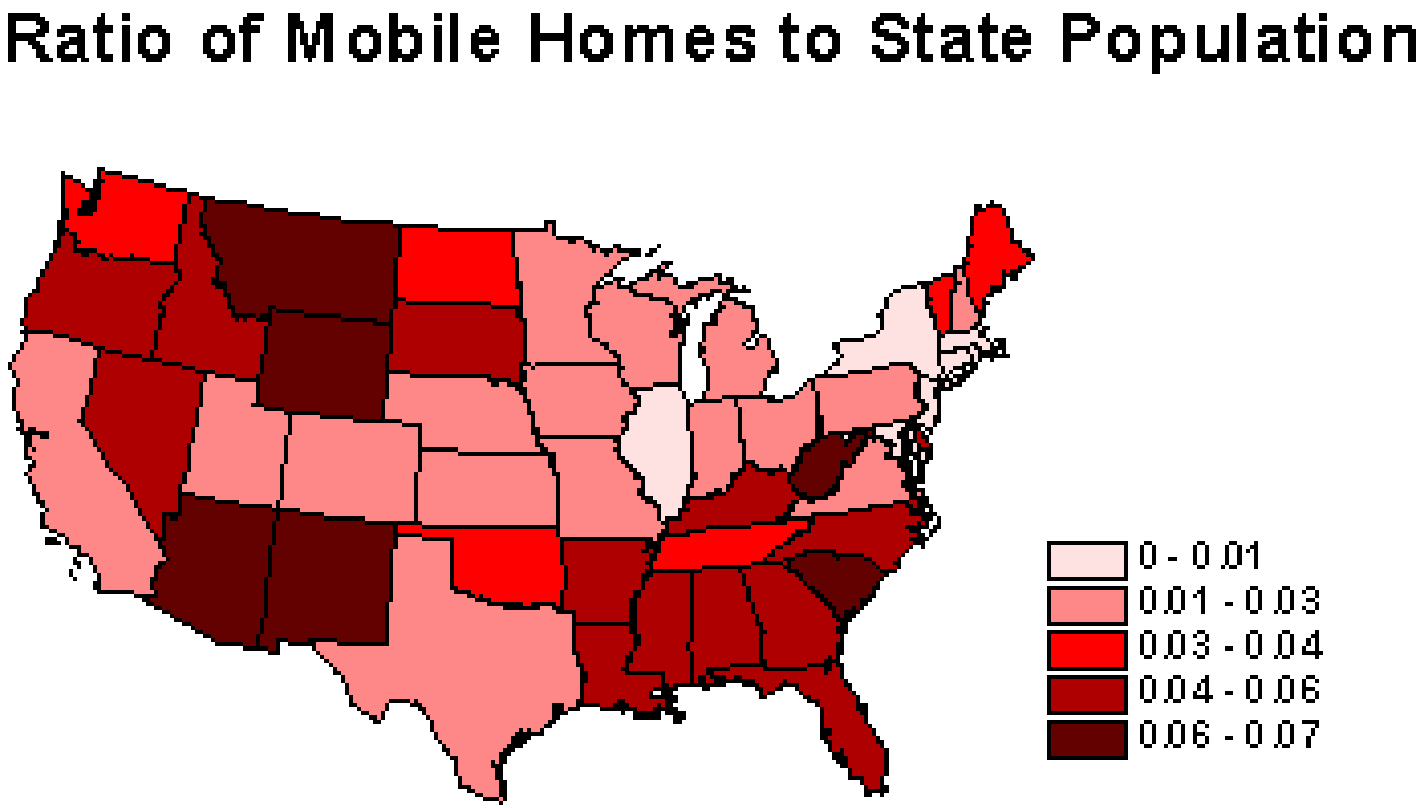
Higher than Average

Lower than Average

Data source: National Cancer Institute

Choropleth

- Data values are generally **classified** into ranges

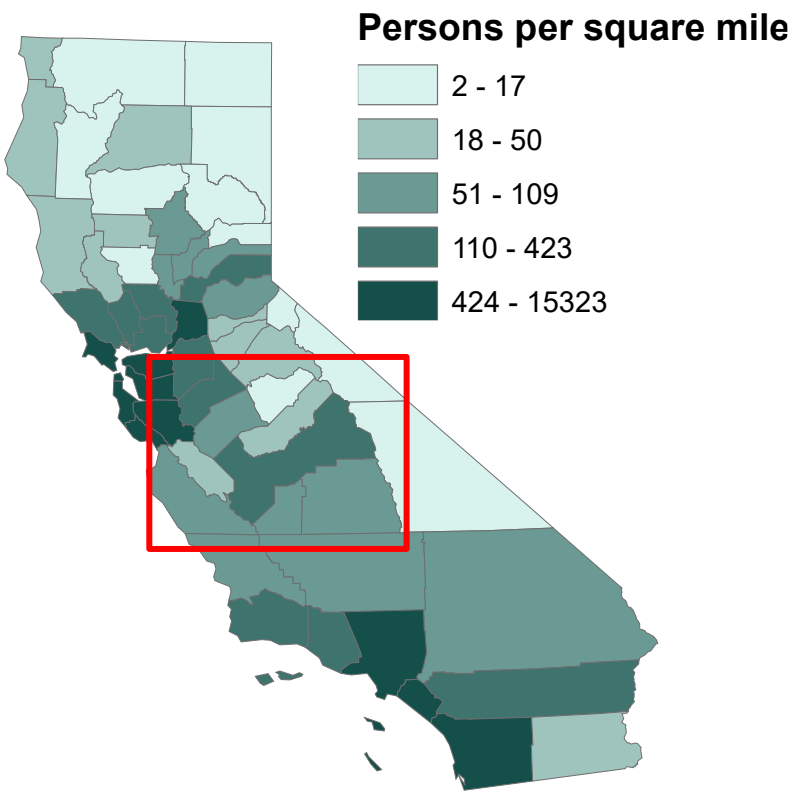


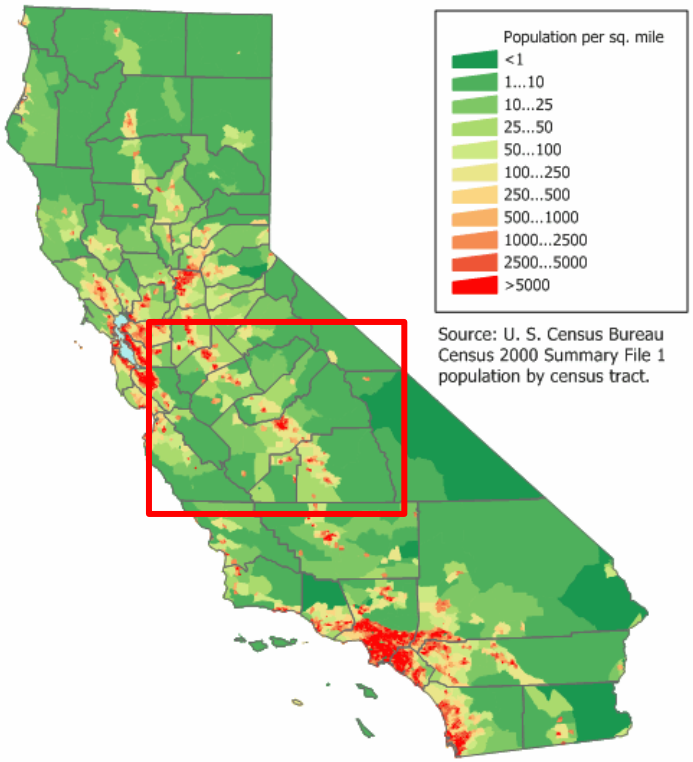
Choropleth: Challenges

- Easy to create a population map
- Polygons can produce misleading impressions
- Modifiable Area Unit Problem:
  - Impression that mapped phenomena are **uniformly** spatially distributed within each polygon unit
  - This is often not true!
- **Boundaries** of enumeration units are frequently **unrelated** to the spatial distribution of the phenomena being mapped
- Larger polygons dominate visually

A scale of analysis problem

Distribution of phenomena (discrete data)



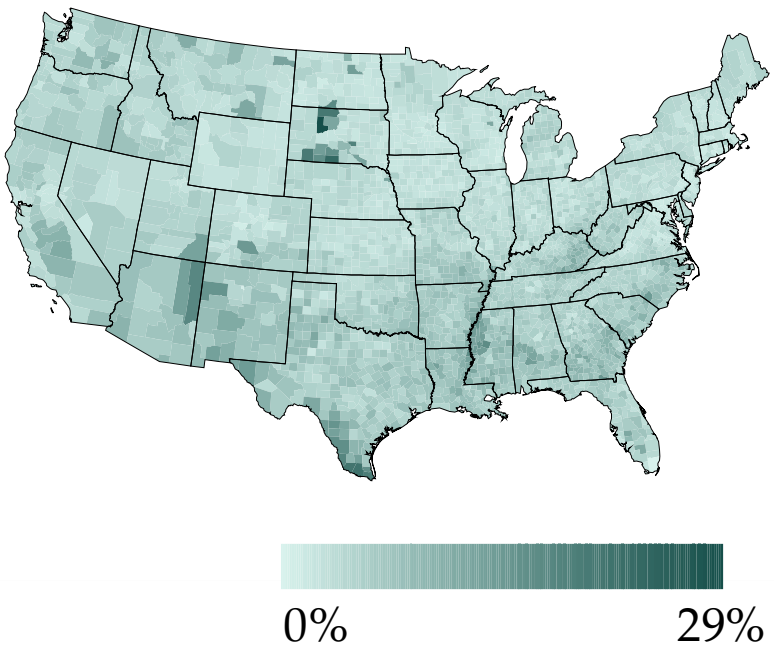


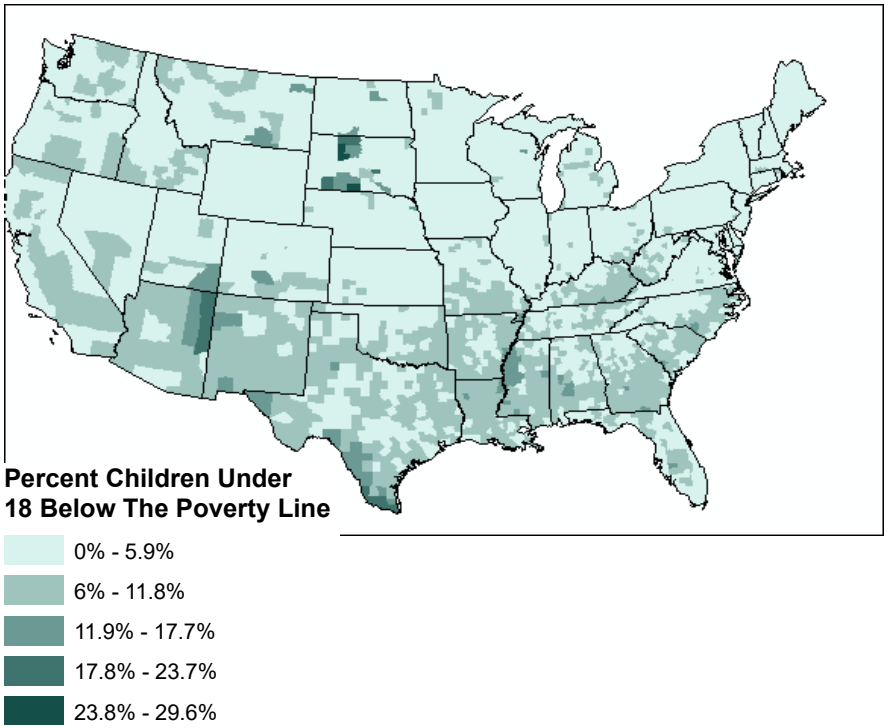
Do we need to bin our data?

A common cartographic debate

Classified (classed) vs. unclassified (un-classed)

- Classified → data in groups, symbol for group
- Unclassified → raw values, symbols show true value



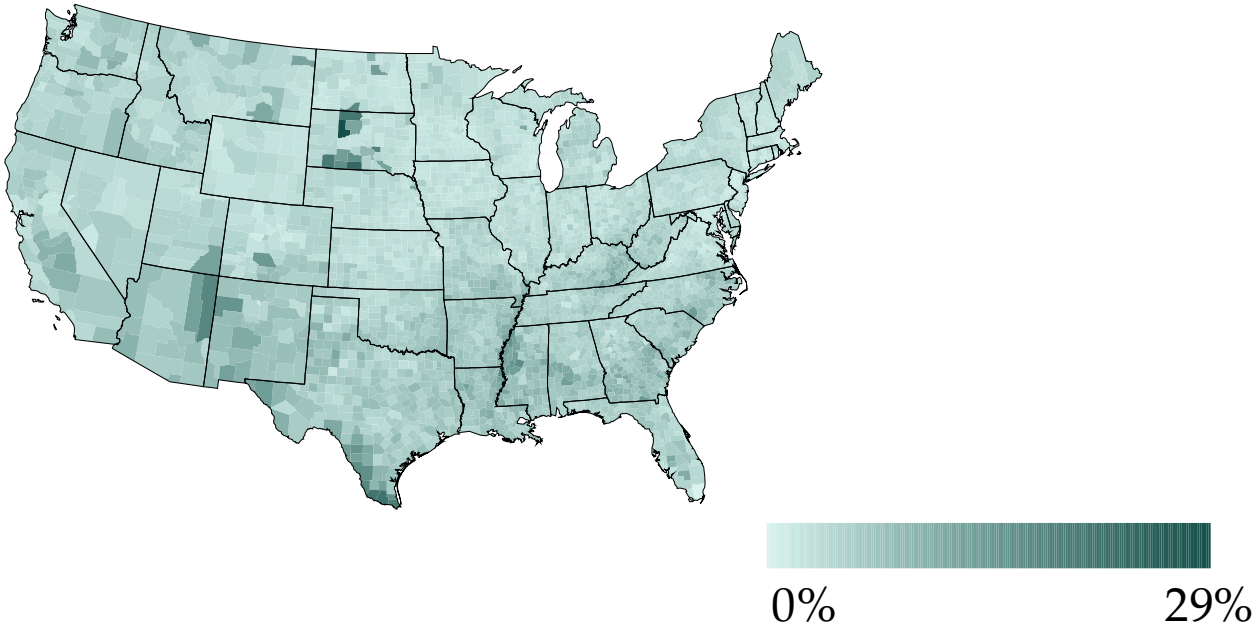


Arguments for unclassed maps

Shows true values

- Enumeration units aren't artificially forced into classes of "like values"

Reader can make his/her own *perceptual* classes



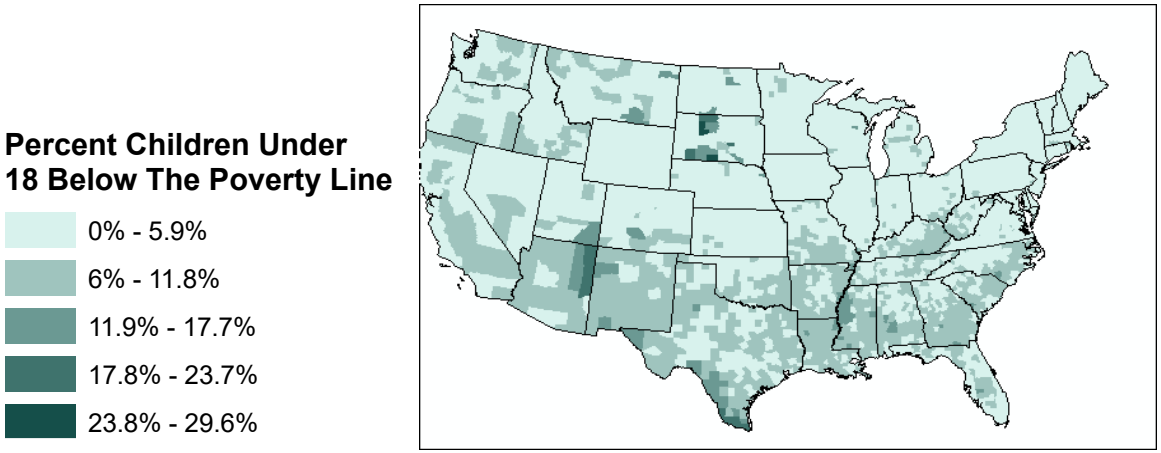
Arguments for classified maps

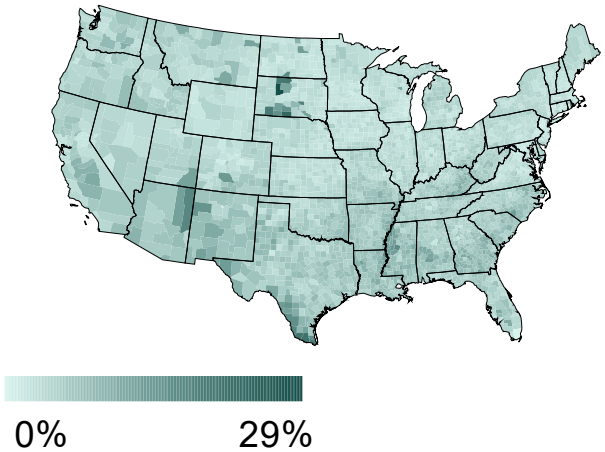
It's easier to *make* a classified map

- True until relatively recently

It's easier to *read* a classified map

- Easier to discriminate between a small number of symbols (5 +/- 2?)





Making a classed map

Select appropriate visual variable

Select number of class breaks

- 3 – 9 is optimal range
- Smaller number = more simplified pattern, less variation

Select appropriate classification method

- Manual
- Statistical
- Combination of both

Manual classification methods

- Exogeneous characteristics
  - Values outside the data distribution

Household income

100k +

50k-99k

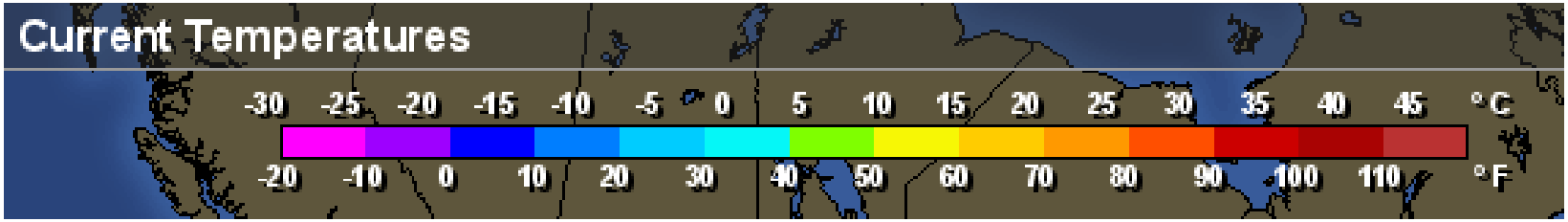
25k-49k

13.7k-24k

Poverty line
$13,690
(2 persons)

Under 13.6k

- Arbitrary values
  - Class breaks based on "nice" numbers

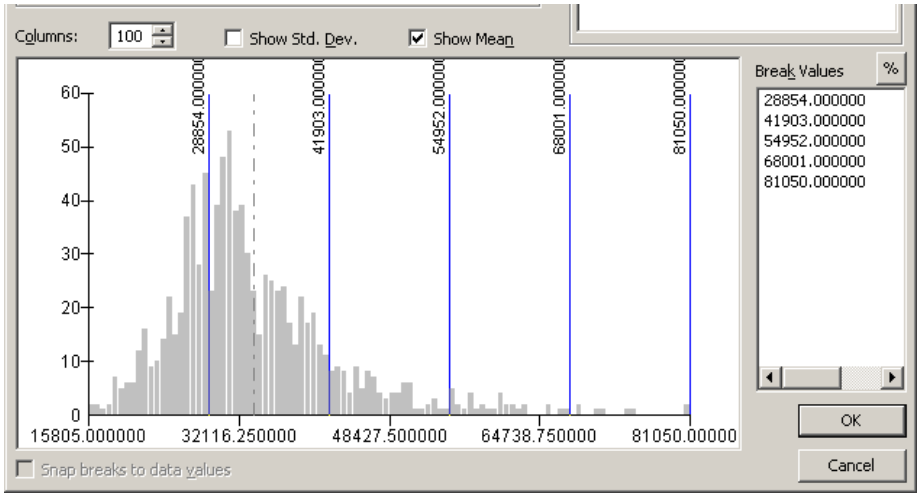


Classification methods

Statistical

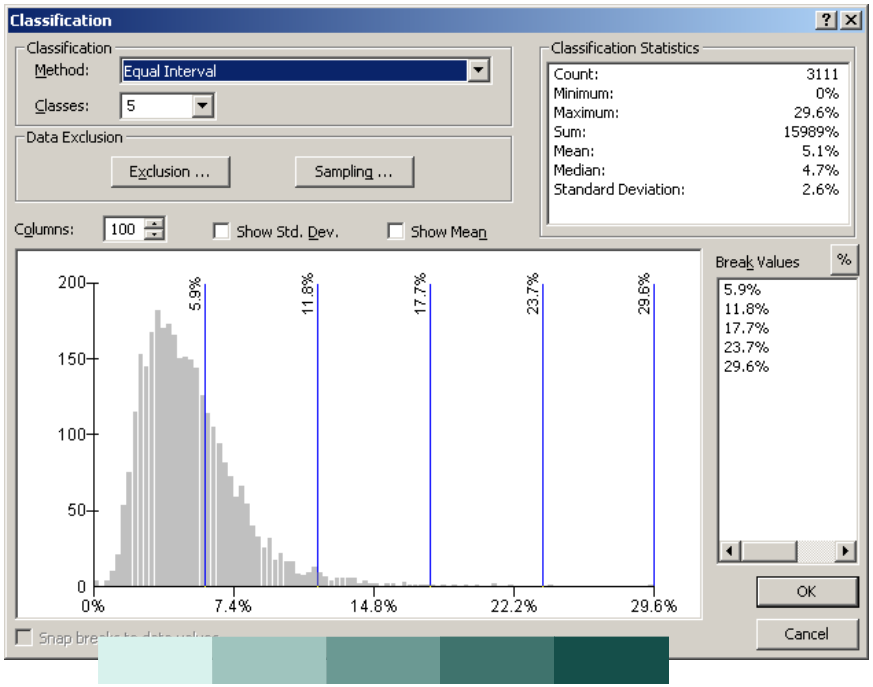
- Equal intervals
- Quantiles
- Mean – Standard deviation
- Natural breaks (Jenks)

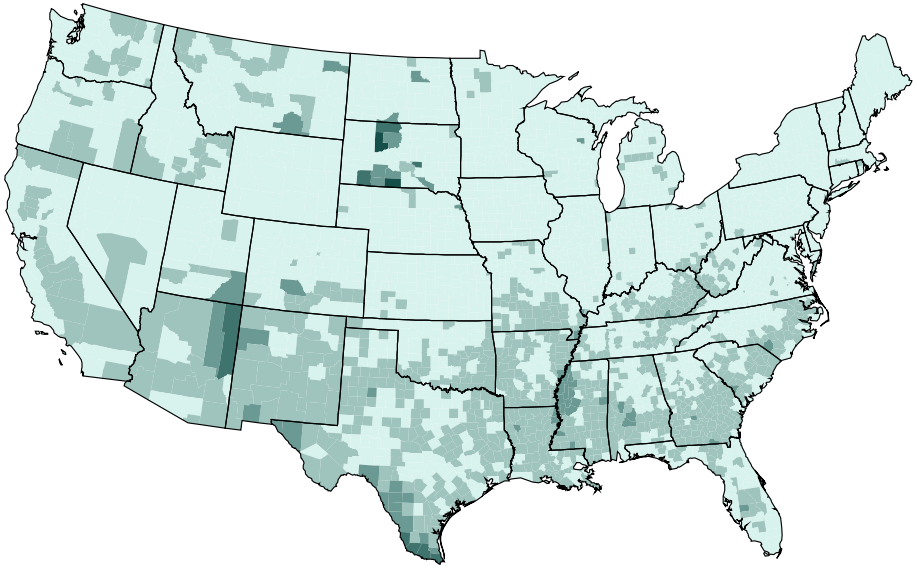
Based on the data distribution



Equal interval in action

- Percent population under 18 living below poverty line

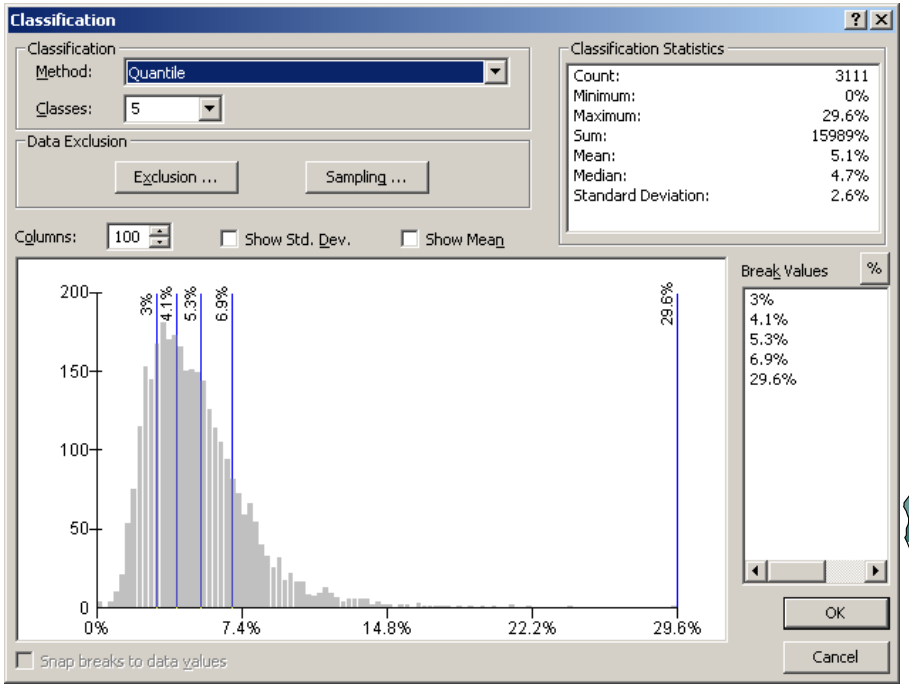




Note the empty or almost empty classes

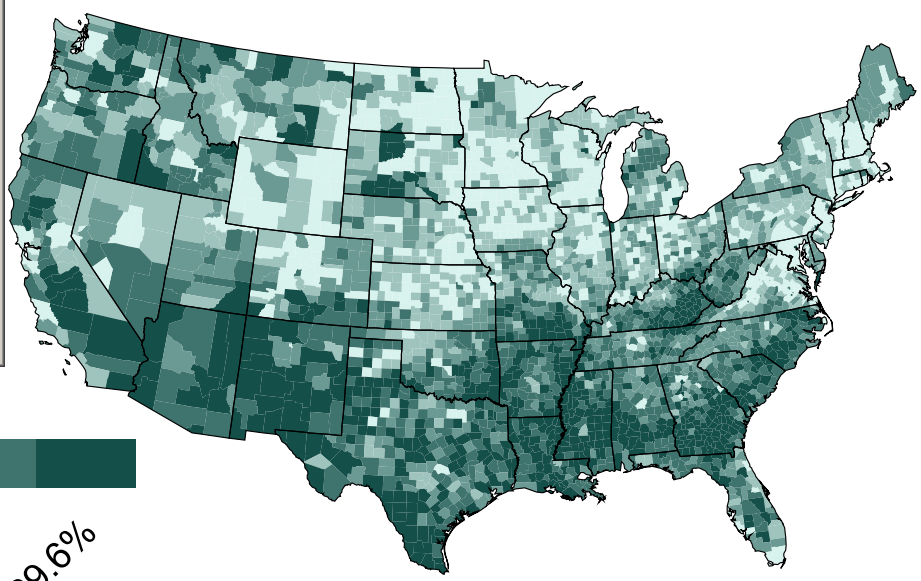
Quantiles in action

- Percent population under 18 living below poverty line



Same number of enumeration units in each class

Classes different sizes; highest class has huge range



0% - 3%

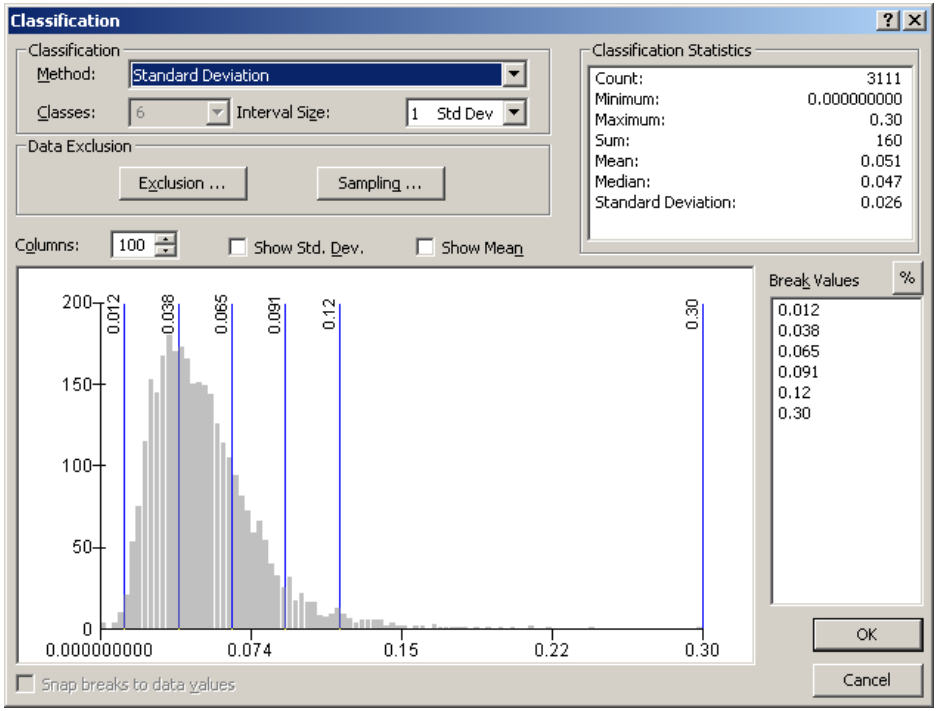
3.1% - 4.1%

4.2% - 5.3%

5.4% - 6.9%

7% - 29.6%

Mean / Standard Deviation in action



< -1.5 Std. Dev.

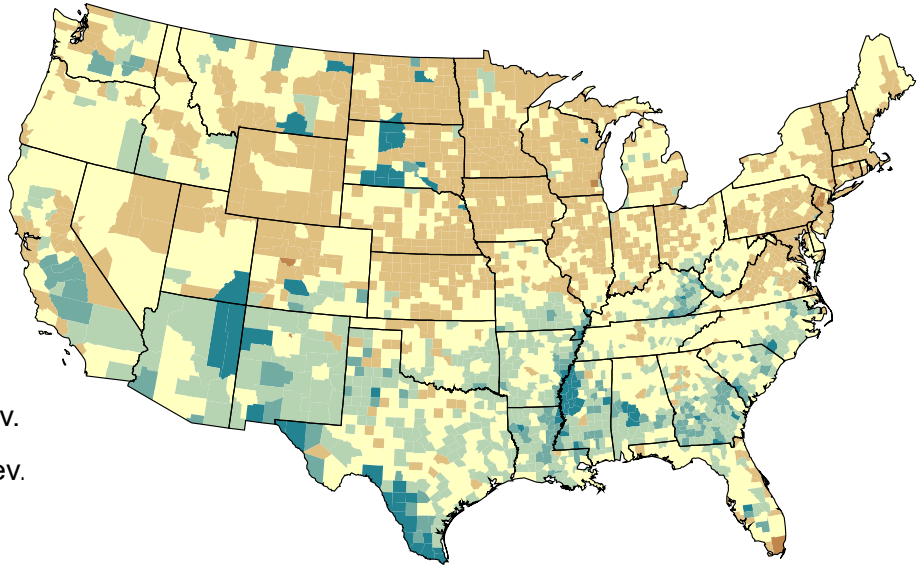
-1.5 - -0.50 Std. Dev.

-0.50 - 0.50 Std. Dev.

0.50 - 1.5 Std. Dev.

1.5 - 2.5 Std. Dev.

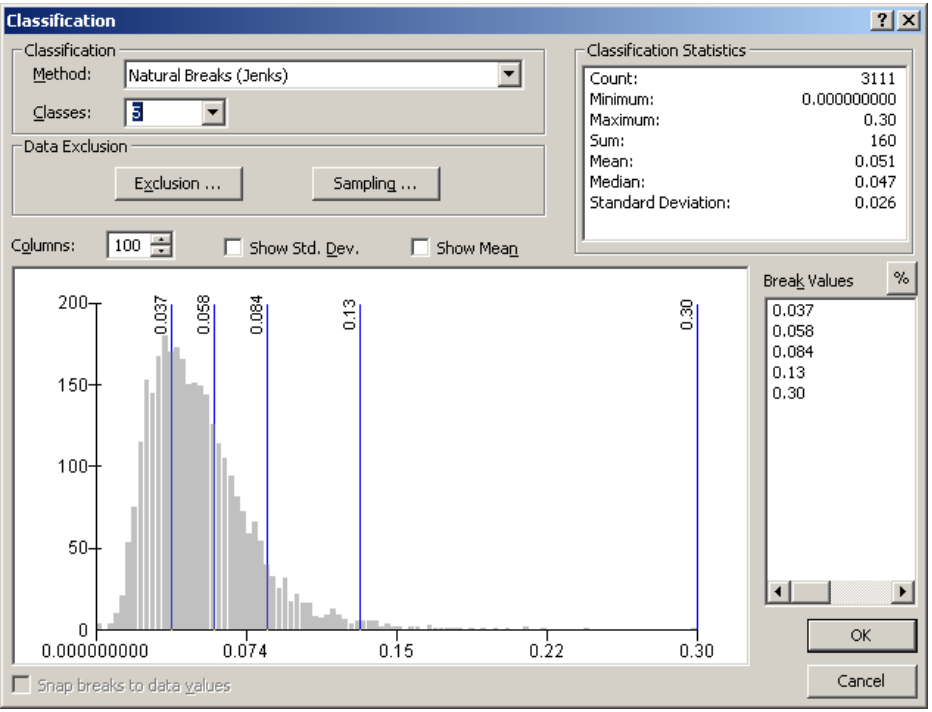
> 2.5 Std. Dev.



Note the diverging color scheme

Natural Breaks (Jenks) in action

A GIS default.



Not a replacement for actually knowing your data!

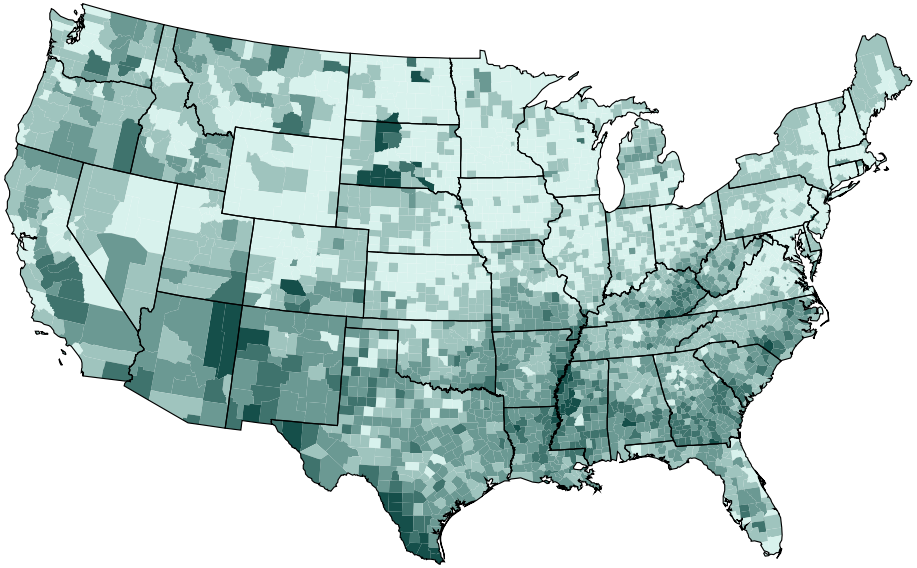
0% - 3.7%

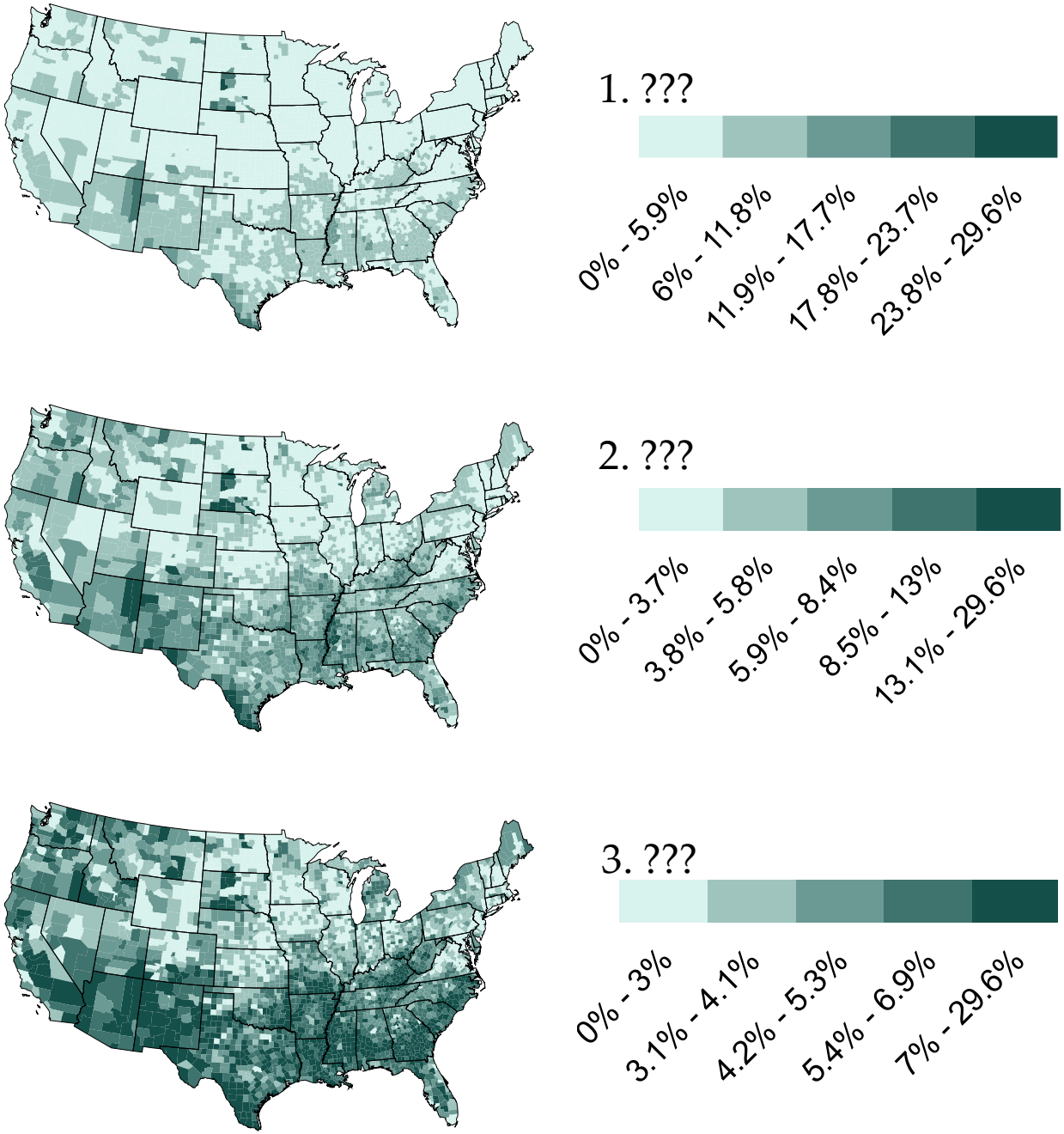
3.8% - 5.8%

5.9% - 8.4%

8.5% - 13%

13.1% - 29.6%




1. ???
0% - 5.9%
6% - 11.8%
11.9% - 17.7%
17.8% - 23.7%
23.8% - 29.6%
2. ???
0% - 3.7%
3.8% - 5.8%
5.9% - 8.4%
8.5% - 13%
13.1% - 29.6%
3. ???
0% - 3%
3.1% - 4.1%
4.2% - 5.3%
5.4% - 6.9%
7% - 29.6%

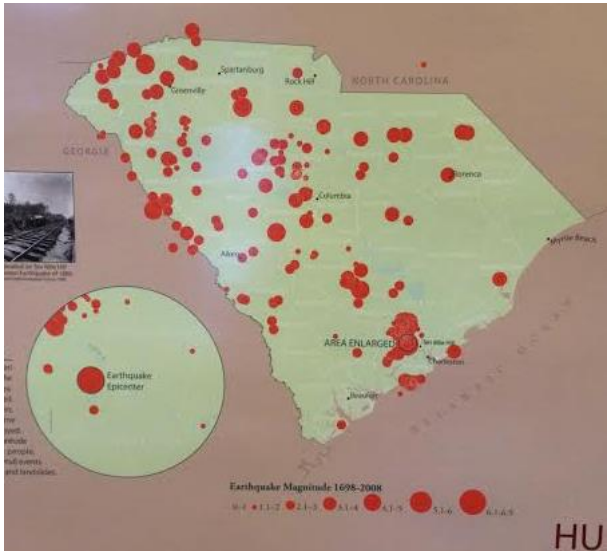
Which legend?

- Over 19,766
- 16,988 – 19,765
- 15,124 – 16,987
- 12,889– 15,123
- 12,345 – 12,888

- Over 20,000
- 17,001 – 20,000
- 15,001 – 17,000
- 13,001 – 15,000
- 12,000 – 13,000

Proportional/Graduated Symbols

- Size of symbol is **proportional** to size of data value
- Frequently used for mapping points' attributes
  - avoids distortions due to area size as seen in choropleth maps



What kind of map?

How does phenomenon change?

abrupt

smooth

discrete

continuous

Where does phenomenon occur?

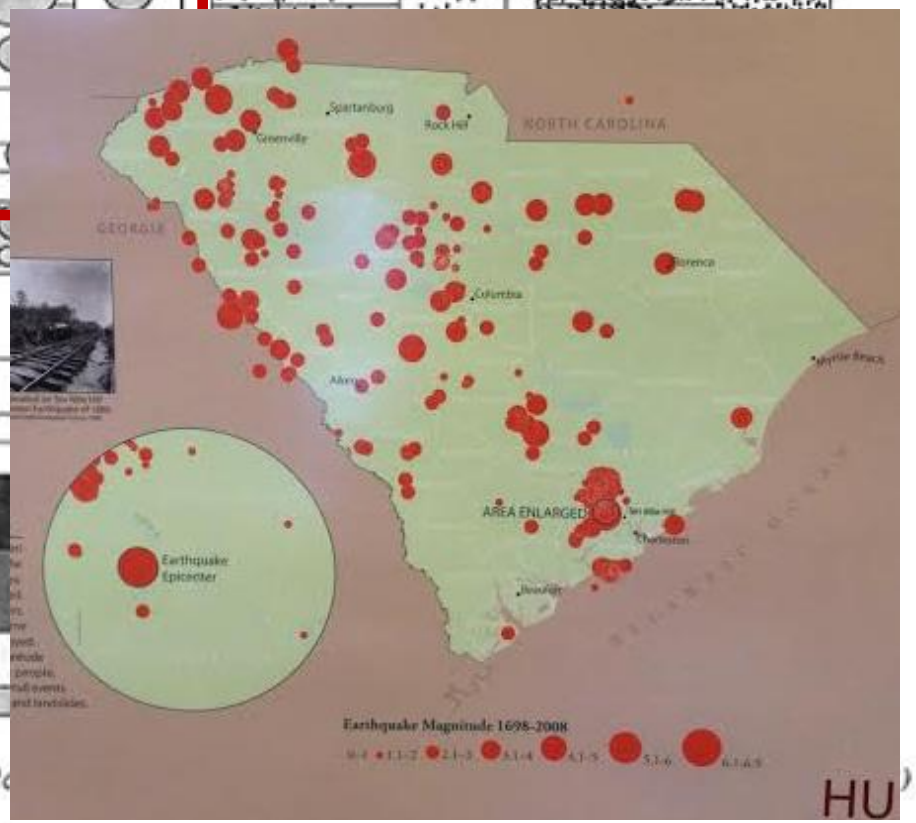
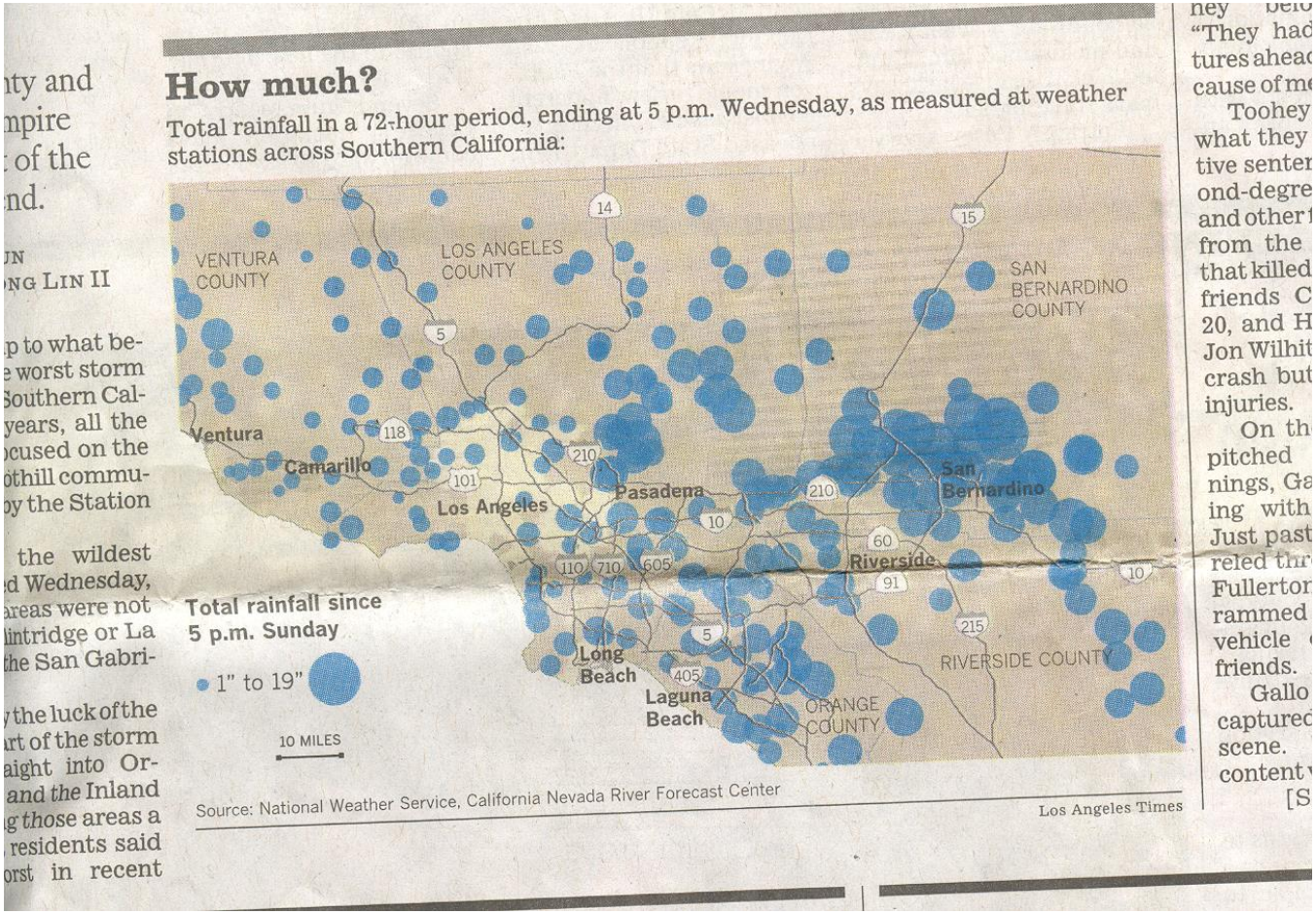


Figure 3.24 Mat

with permission.

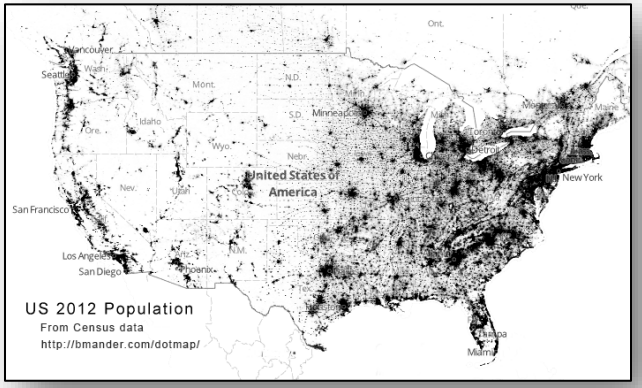
Proportional symbols

Clusters / groups & "independence"



Dot Density Maps

- Dot density maps provide an immediate picture of density over area
- 1 dot = some quantity of data value
  - e.g. 1 dot = 500 persons, generally associated with polygon unit
- **Placement** of dots within polygon enumeration units can be an issue, especially with sparse data



What kind of map?

How does phenomenon change?

Where does phenomenon occur?

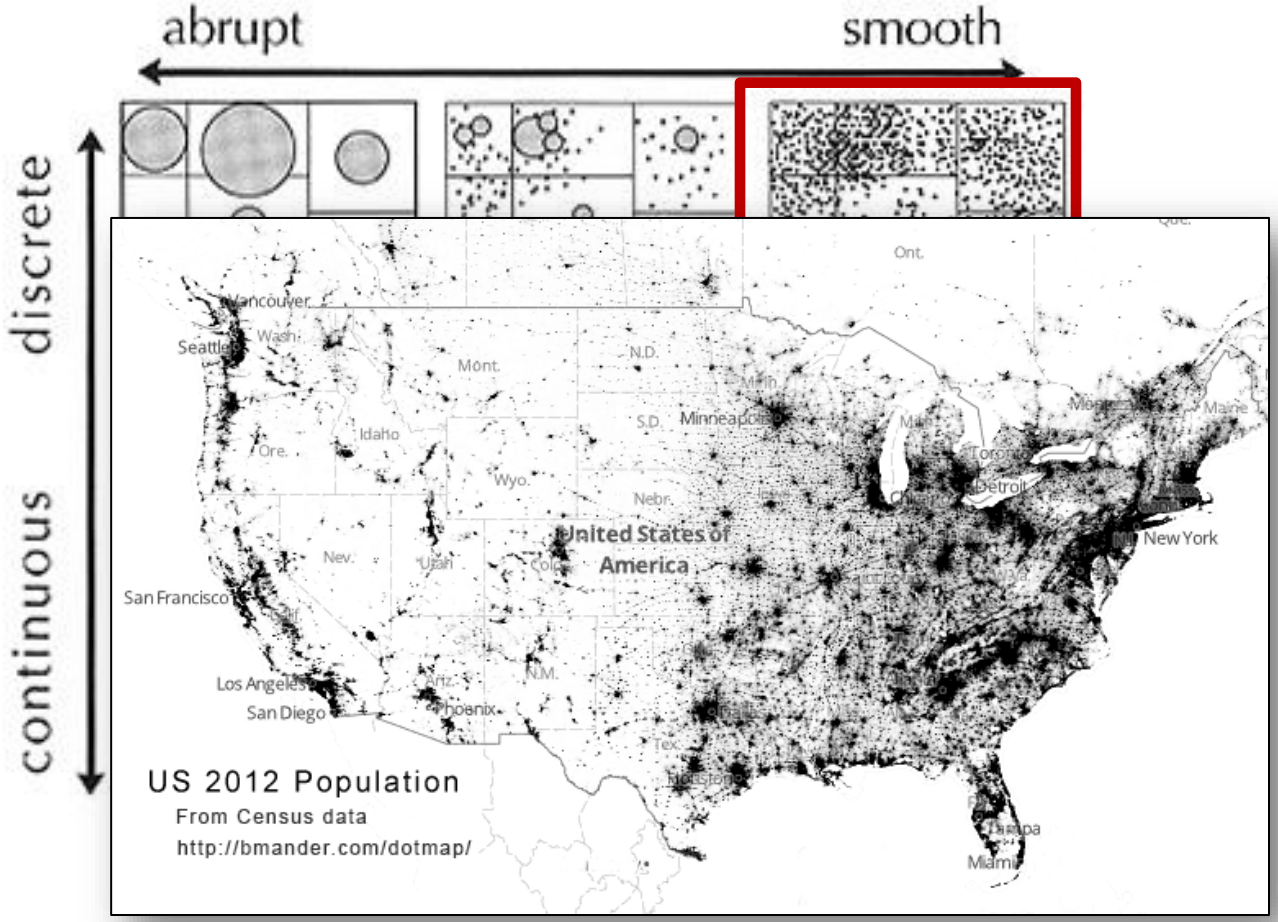


Figure 3.24 Matching map types to data models. Reproduced from MacEachren (1992) with permission.

Dot mapping guidelines

Figure out where the phenomena being mapped can really be located

- Are there constraints (e.g,. Uninhabitable areas)
- What do you know about the real-world distribution of the phenomena

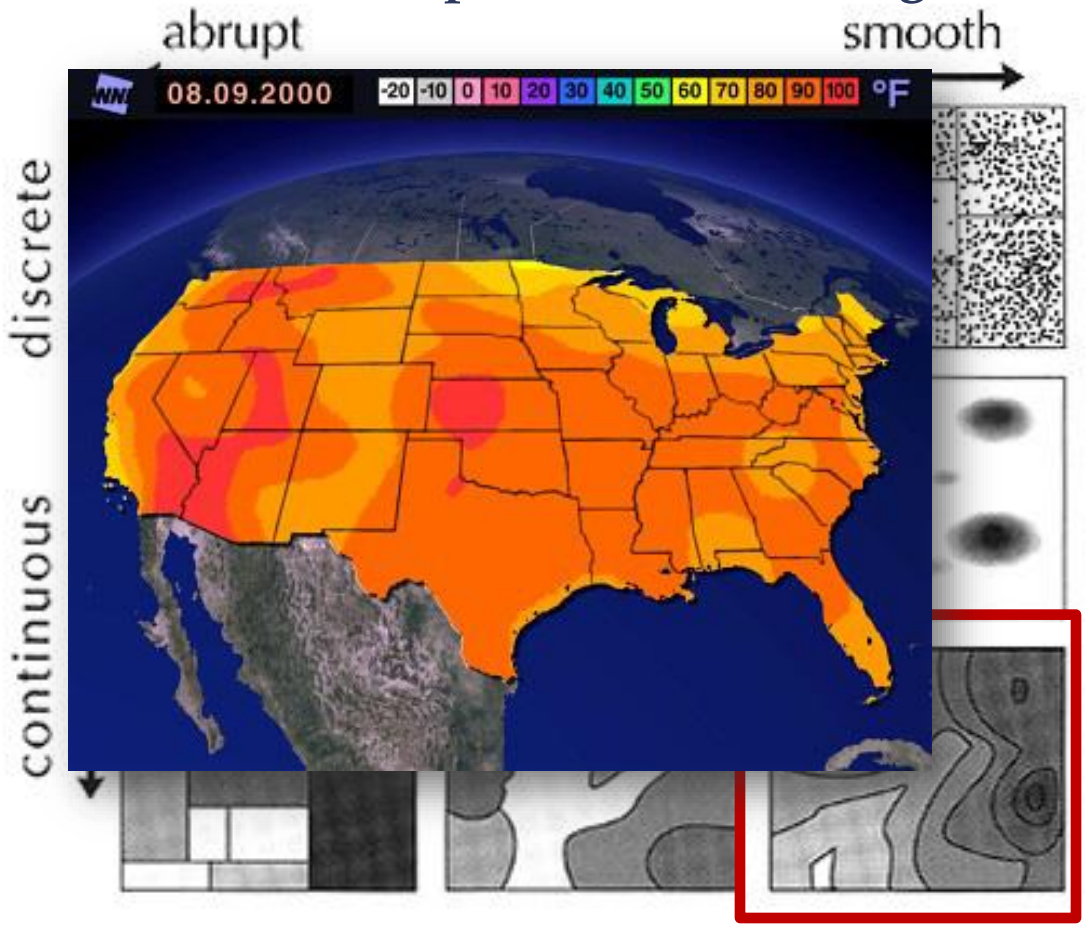
Use the smallest enumeration units possible for your map

- Dots will be placed randomly through the enumeration unit, so pick something with as little room for error as possible

What kind of map?

How does phenomenon change?

Where does phenomenon occur?



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What is wrong with this map?

...

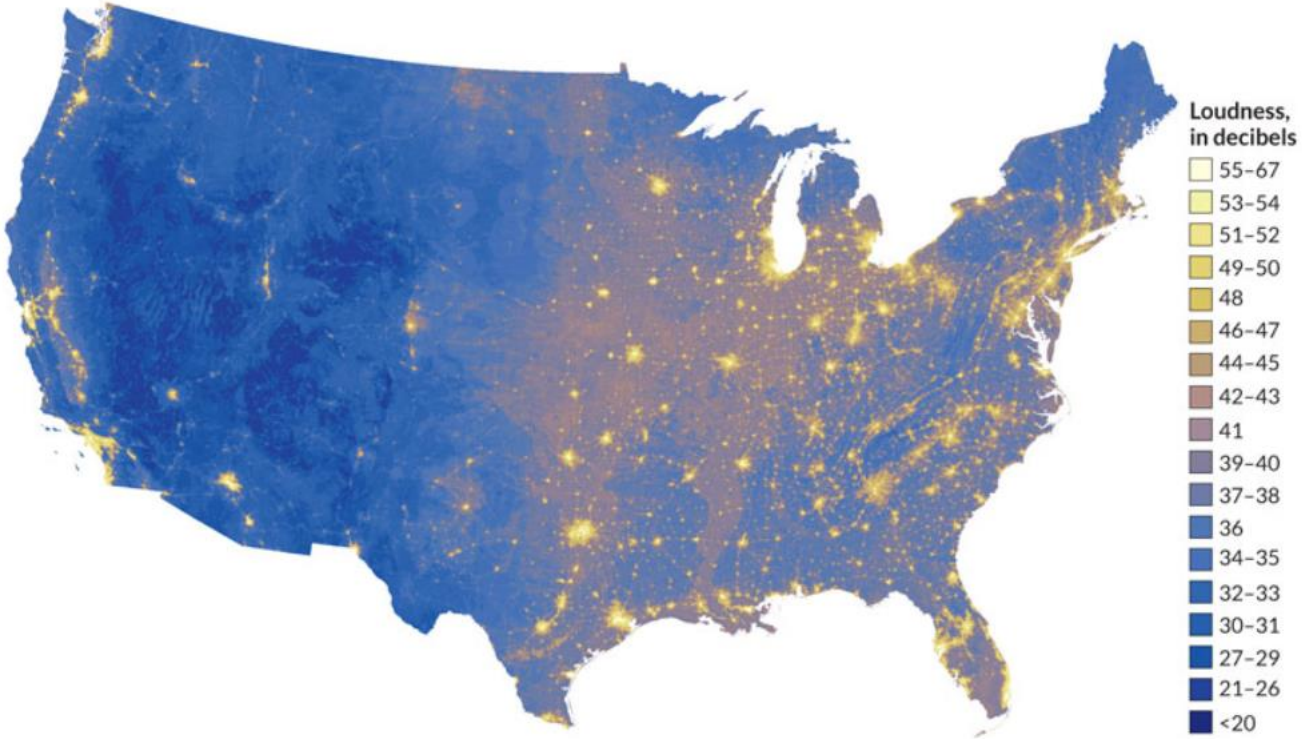
An ambitious National Park Service project exploits computer algorithms to predict the loudness of a typical summer day from coast to coast.

SCIENCE VISUALIZED SUSTAINABILITY, CONSERVATION

A coast-to-coast picture of America's cacophony of sounds

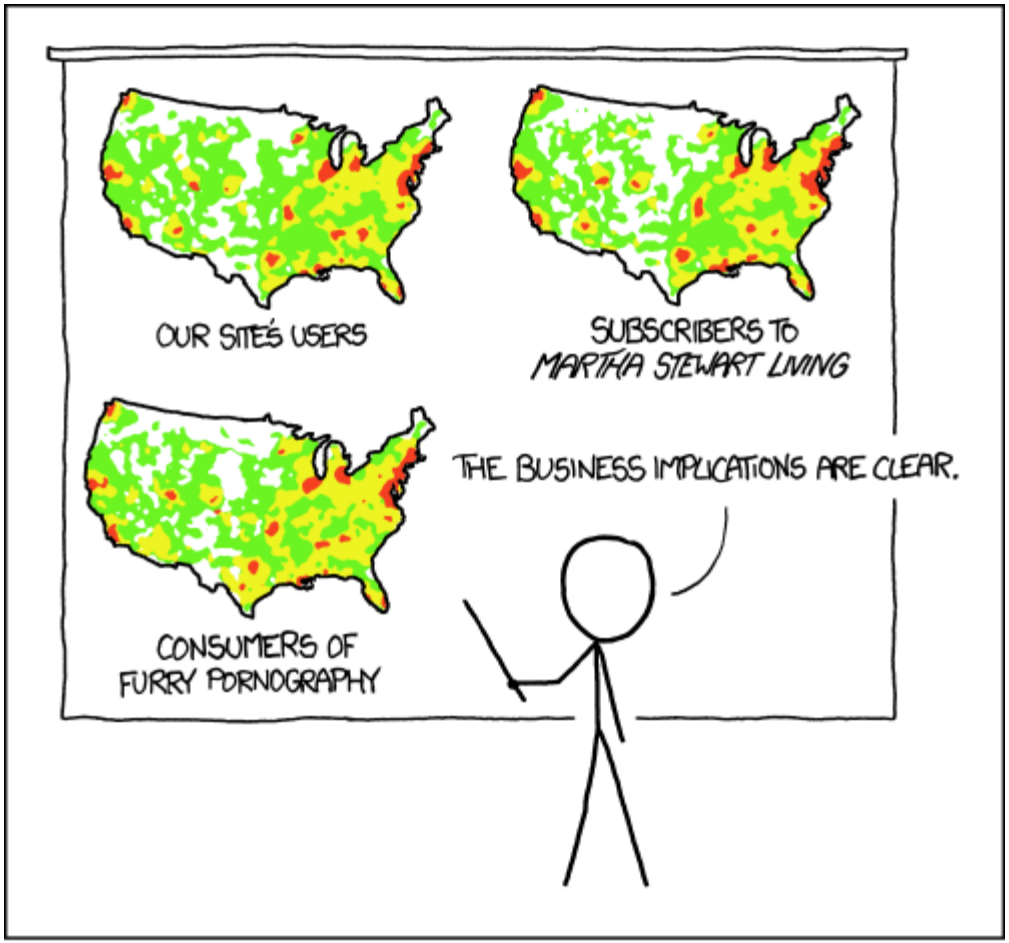
New National Park Service map draws the United States' summer soundscape

BY SUSAN MILIUS 12:12PM, FEBRUARY 16, 2015



SOUNDSCAPE According to a new map created by the National Park Service, the eastern half of the United States is louder than the western United States.

NPS NATURAL SOUNDS & NIGHT SKIES DIVISION



PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE BASICALLY JUST POPULATION MAPS

Source: xkcd.com

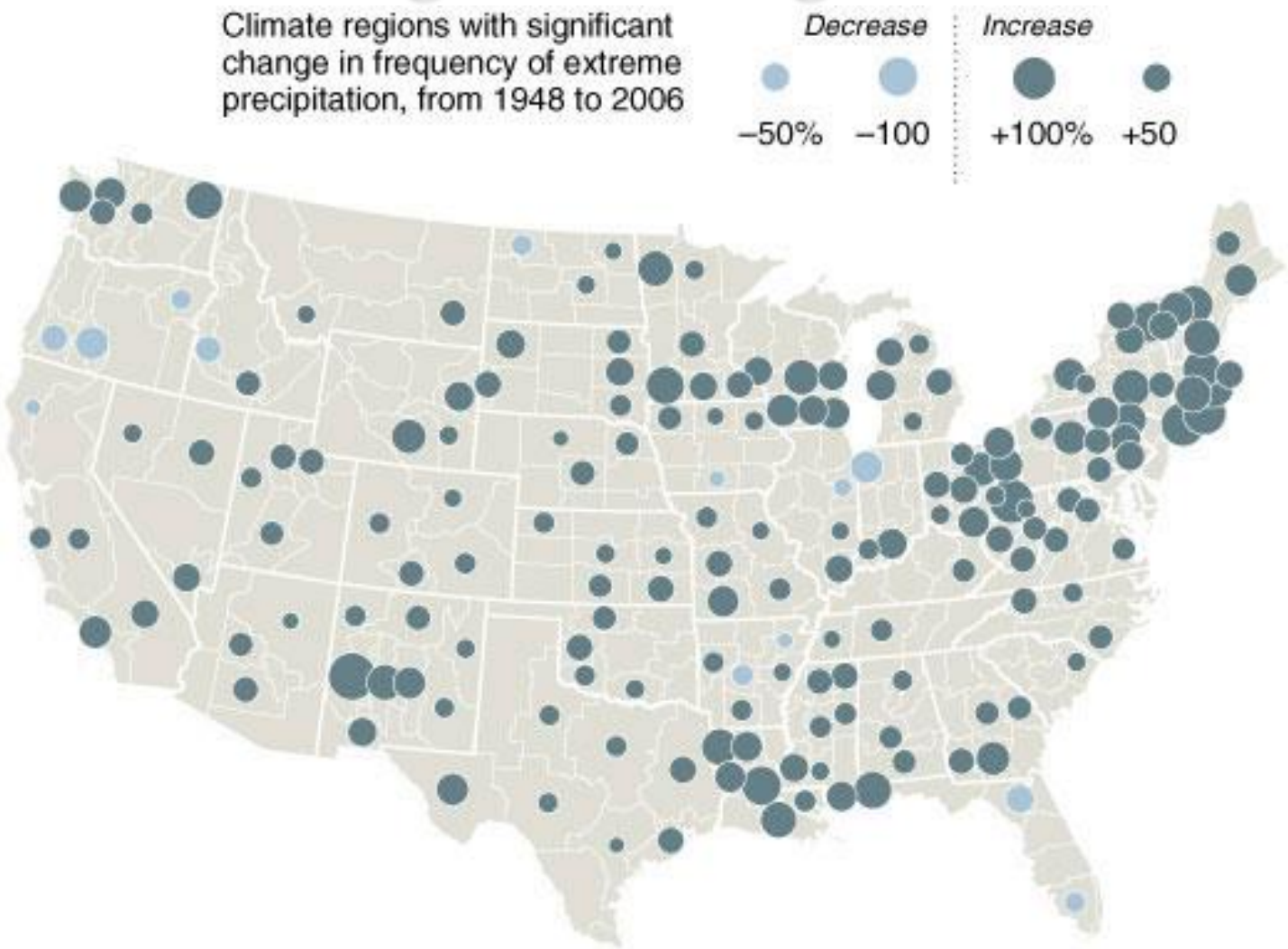
Please, please, please think about color

...

Color – distinguish categories

Where Storms Are Worsening

The number of severe rainstorms and heavy snow has grown significantly in the last 60 years, with some of the largest increases in New England and New York, according to a new report by Environment America, an advocacy group. The trends are consistent with scientists' predictions that global warming is likely to increase the intensity of storms.

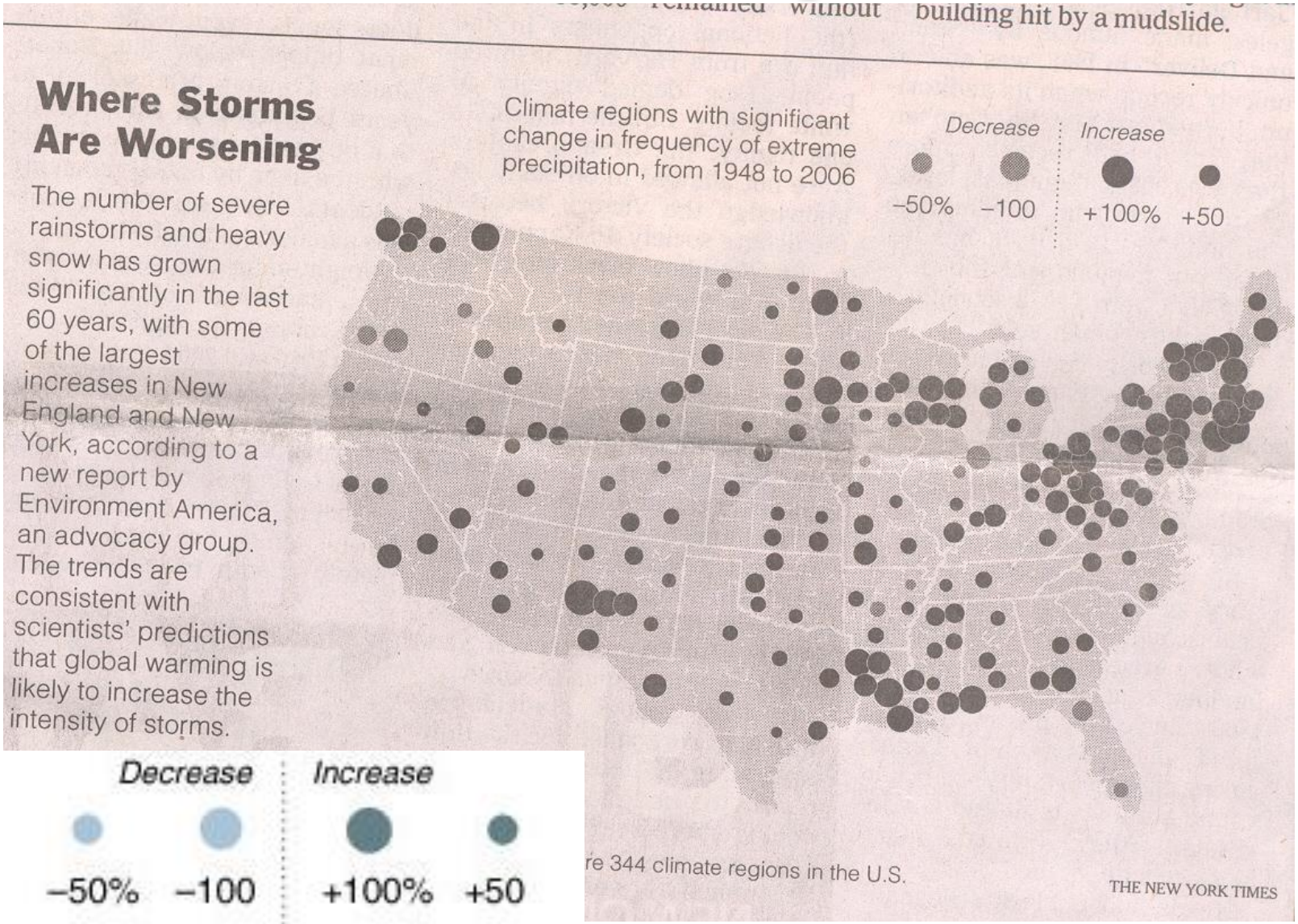


Source: Environment America

Note: There are 344 climate regions in the U.S.

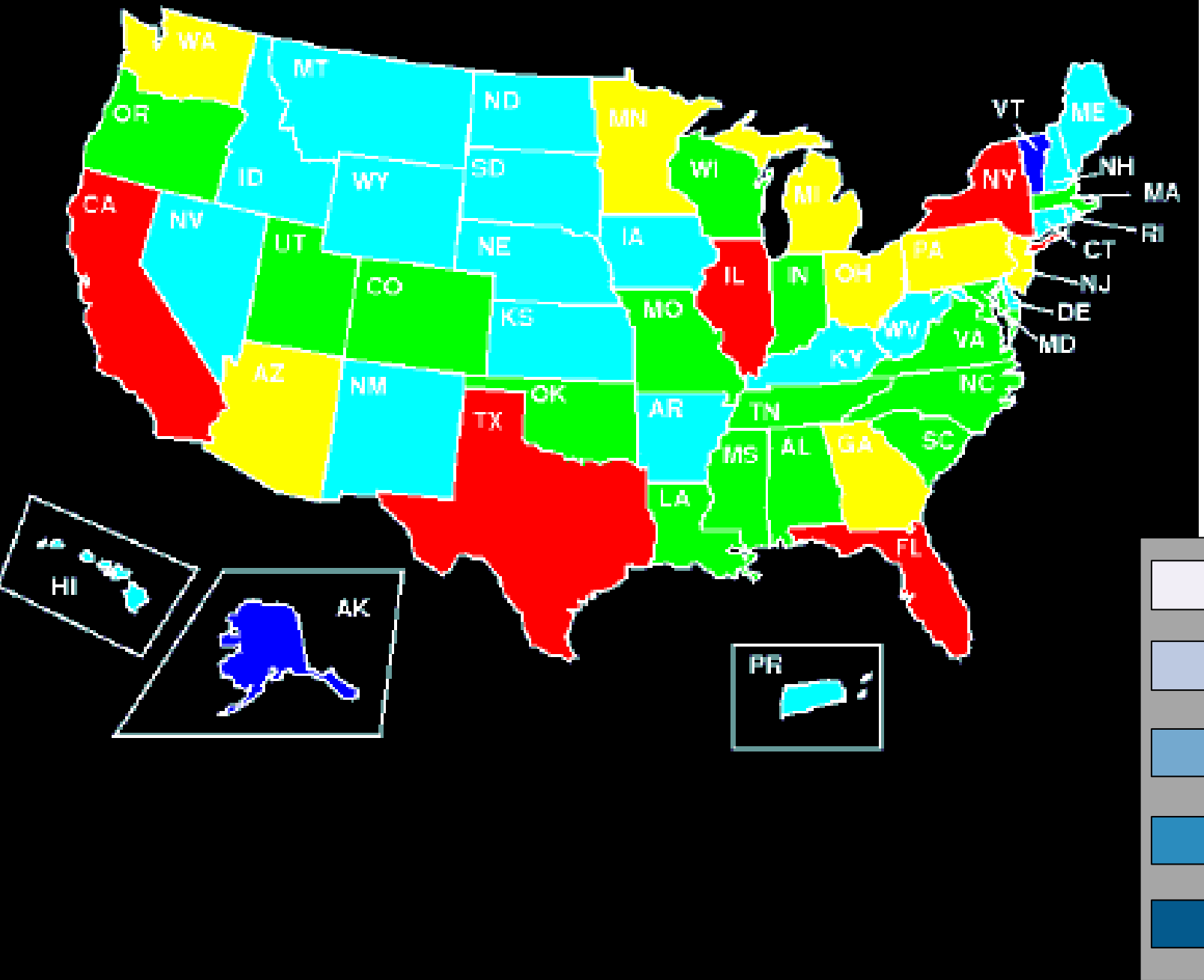
THE NEW YORK TIMES

Color – it's important



Domestic freshwater use (withdrawals and deliveries) for 1990

Source: USGS



- 0-50
- 50-250
- 250-500
- 500-1000
- 1000-4000

It turns out that maps are all sort of wrong

...

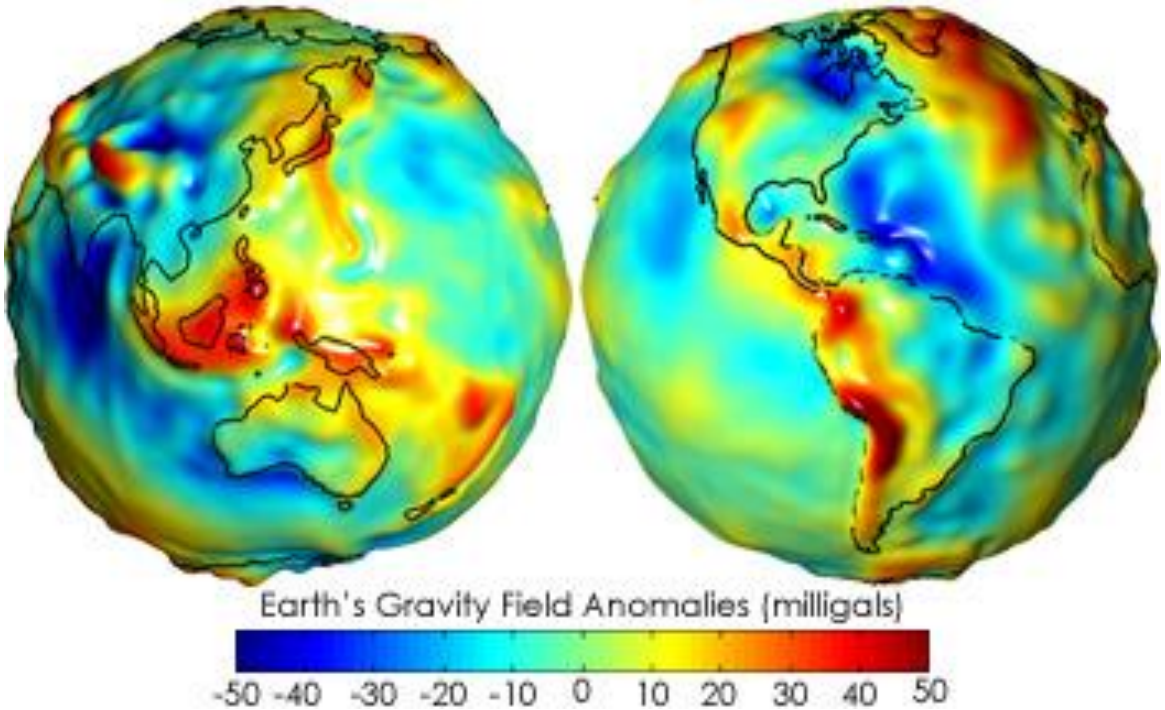
But we can try to make them a little more right

And try not to lie about the parts that actually matter

Map projections

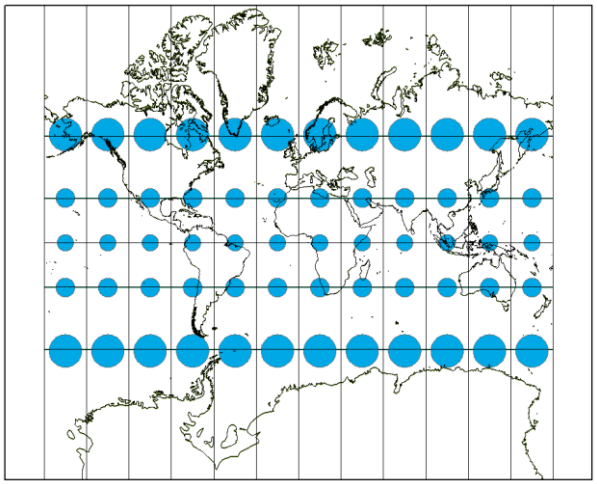
The earth is (sort of) spherical

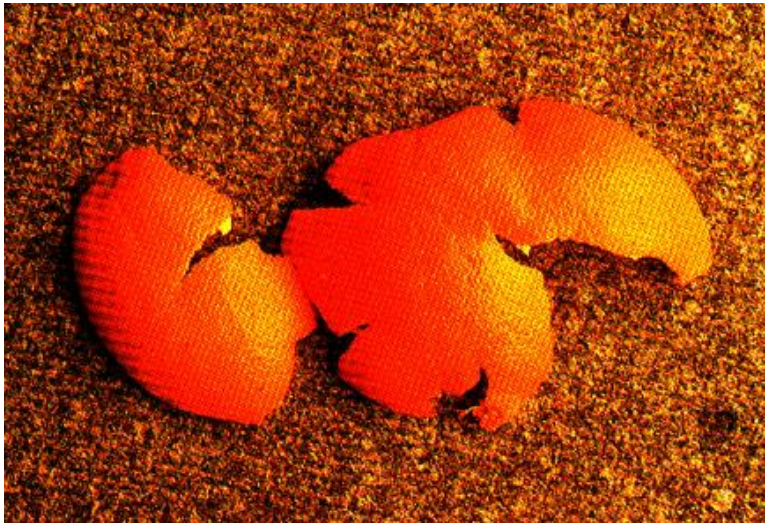
"geoid undulations"



Map projections

Map projections in a nutshell

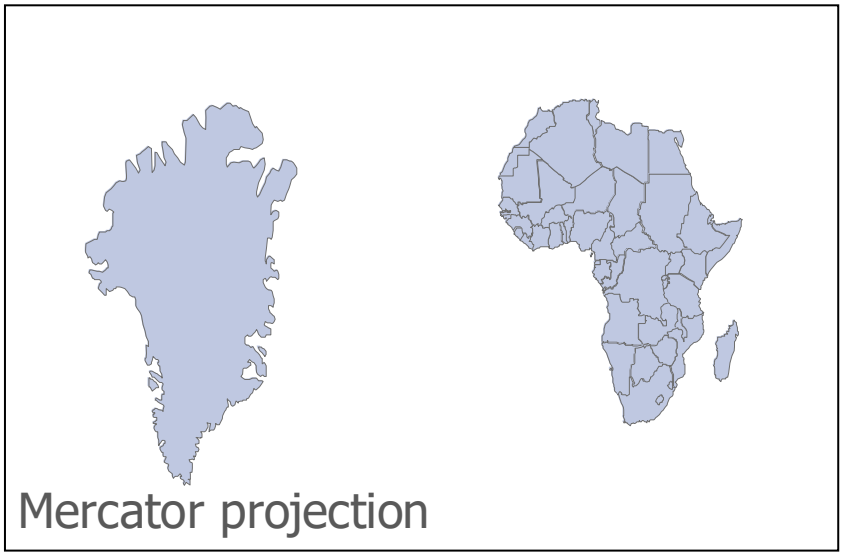


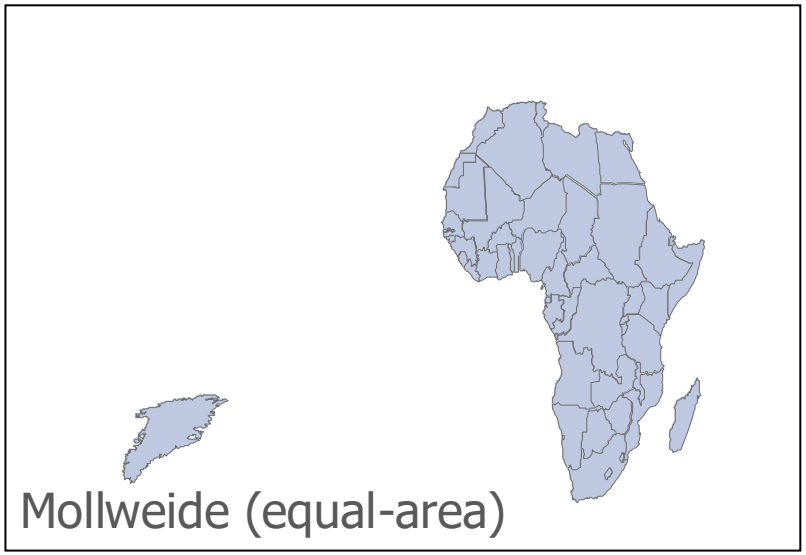


Conformal --- preserves angles
Equal area --- preserves area
Equidistant --- preserves distance
Compromise --- preserves nothing, but generally looks nice

Identify comparison of interest

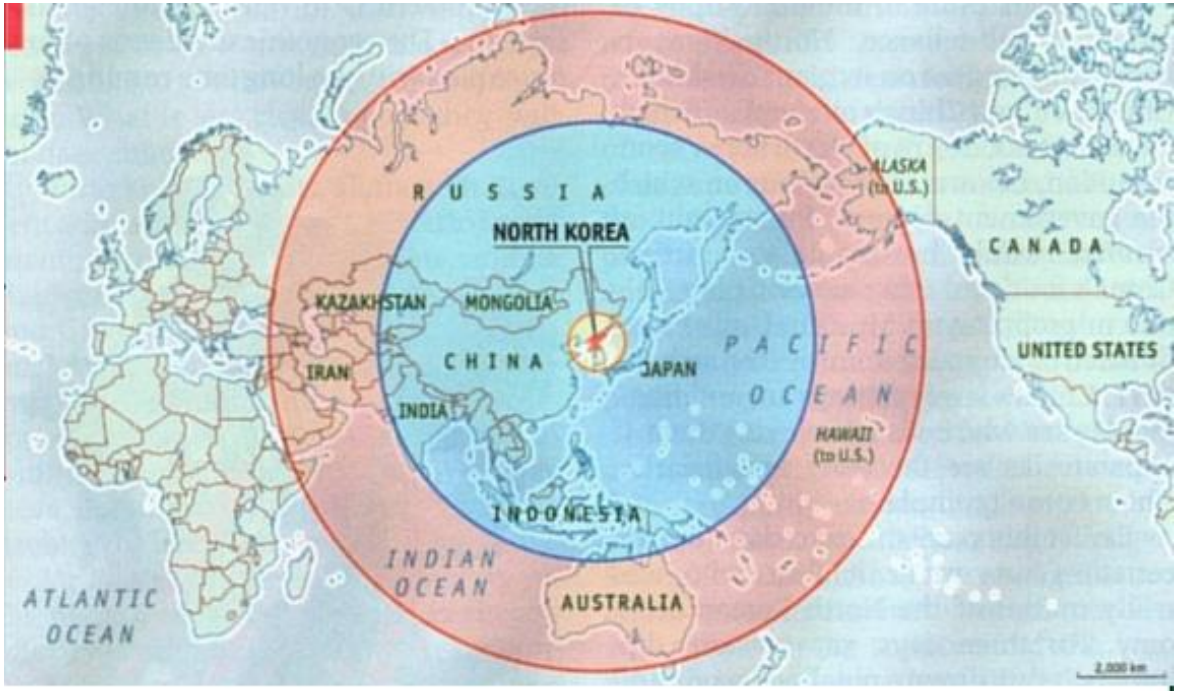
Accuracy will always be compromised





What is important – Areas? Distances? Angles?

Application: The Economist



North Korea's missile threat

Type	Maximum range	Payload	Status
Nodong	1,300 km (810 miles)	700 kg (1,550 pounds)	Currently deployed
Taepodong-1	Up to 10,000 km	Several hundred kg	Test failed 1998, not yet operational
Taepodong-2	10,000–15,000 km	Several hundred kg	Not yet tested

Source: Task Force for US Korea Policy, Centre for International Policy

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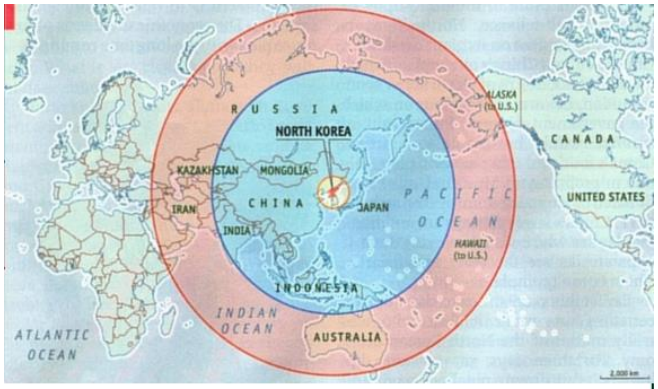

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Source: The Economist

Cartograms

Most thematic maps…

- Minimize distortion of spatial relationships (e.g., area, distances, etc.)

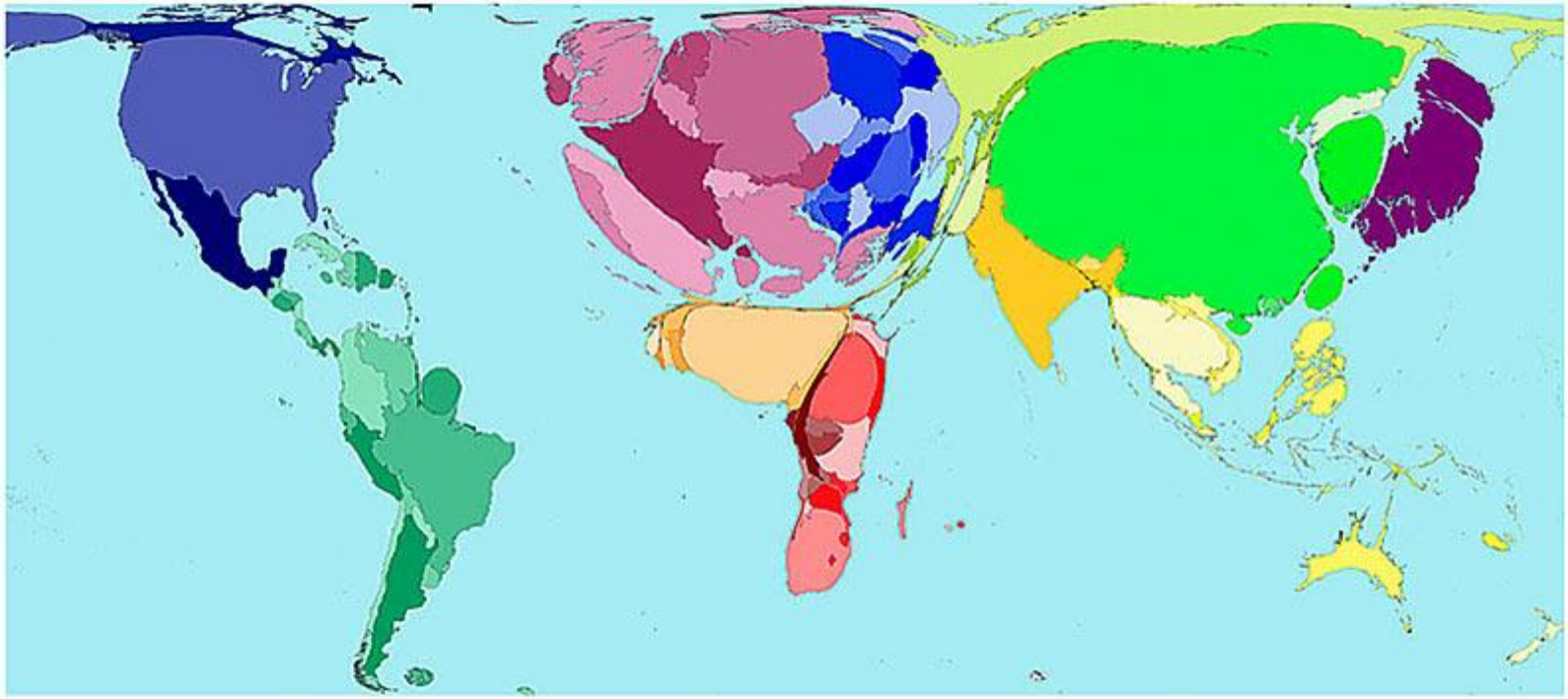
- Represent pattern for a theme using correct locations

- However, size of polygons (choropleth map) can mislead– very densely populated areas are often small, leaving a smaller visual impression, and vice versa.

Cartograms…

- *Intentionally distort spatial relationships* to represent theme

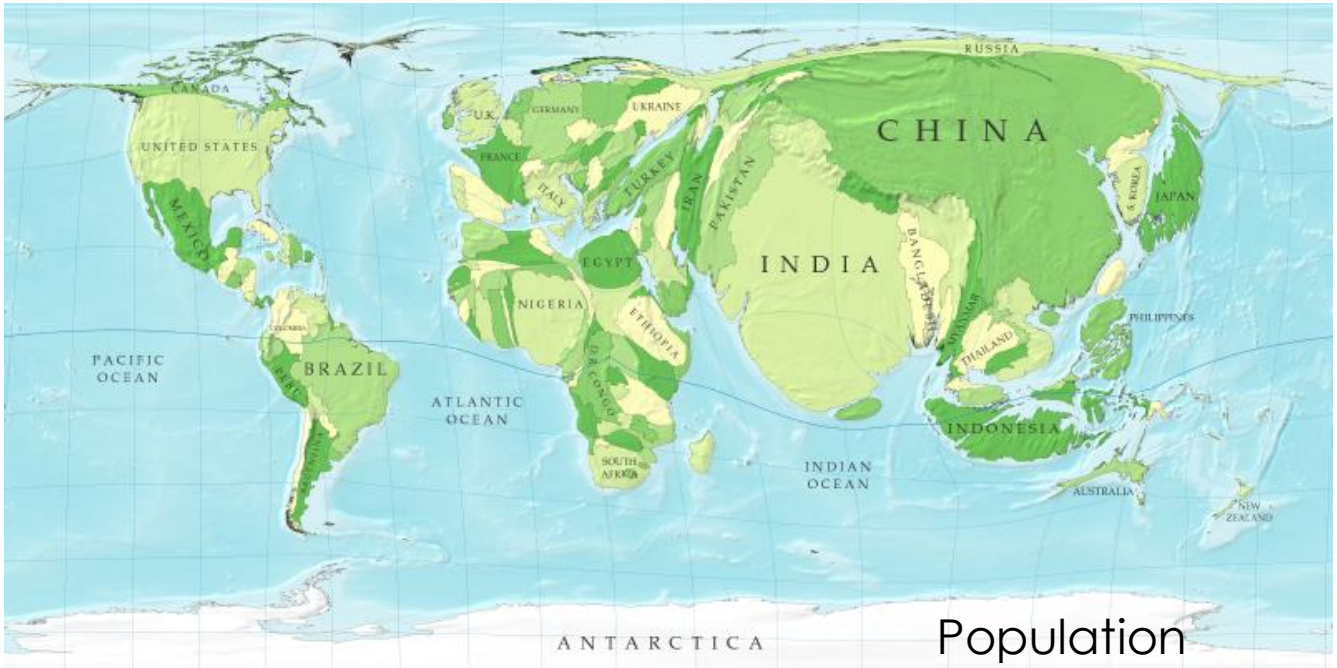
Alcohol consumption



Source: http://img.dailymail.co.uk

Distorting spatial relations

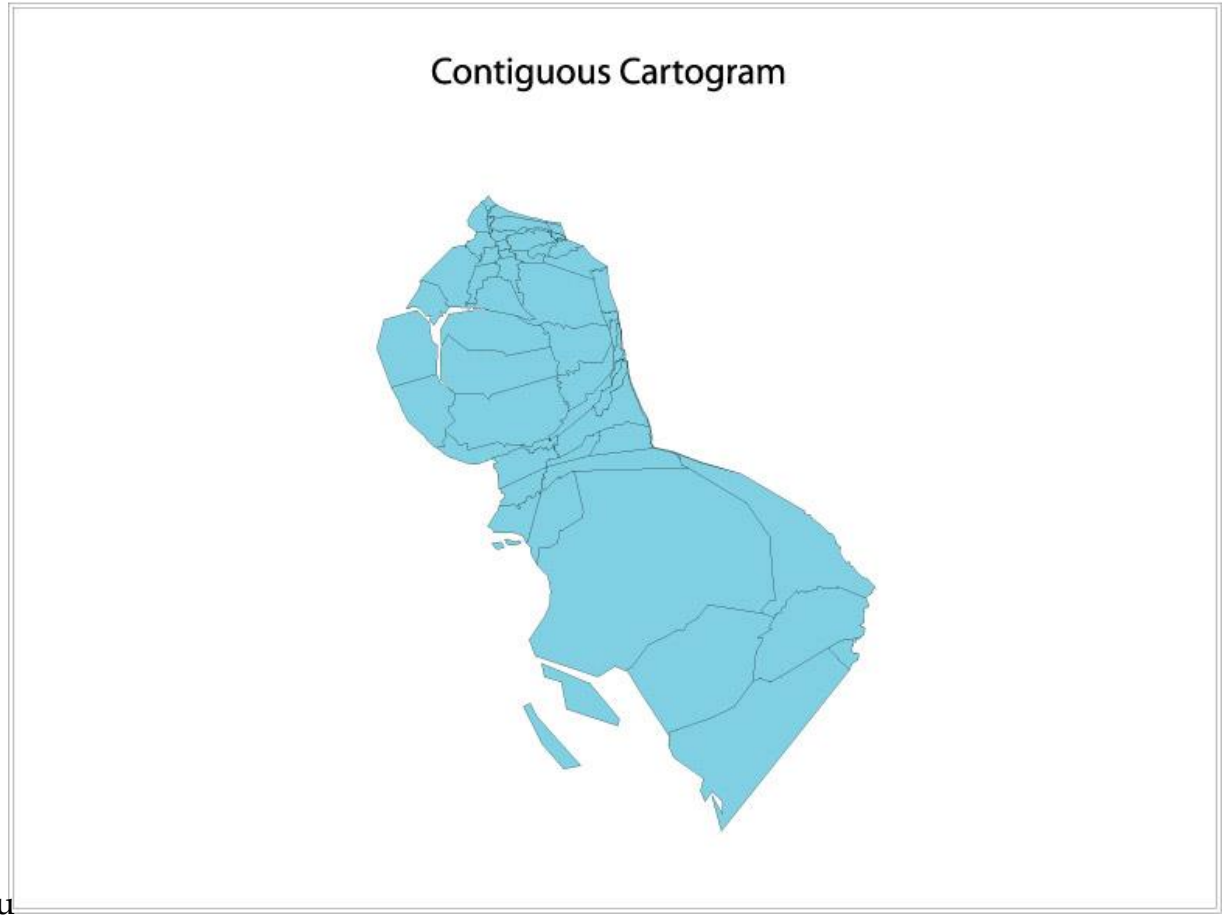
Tends to be used with count data



Population

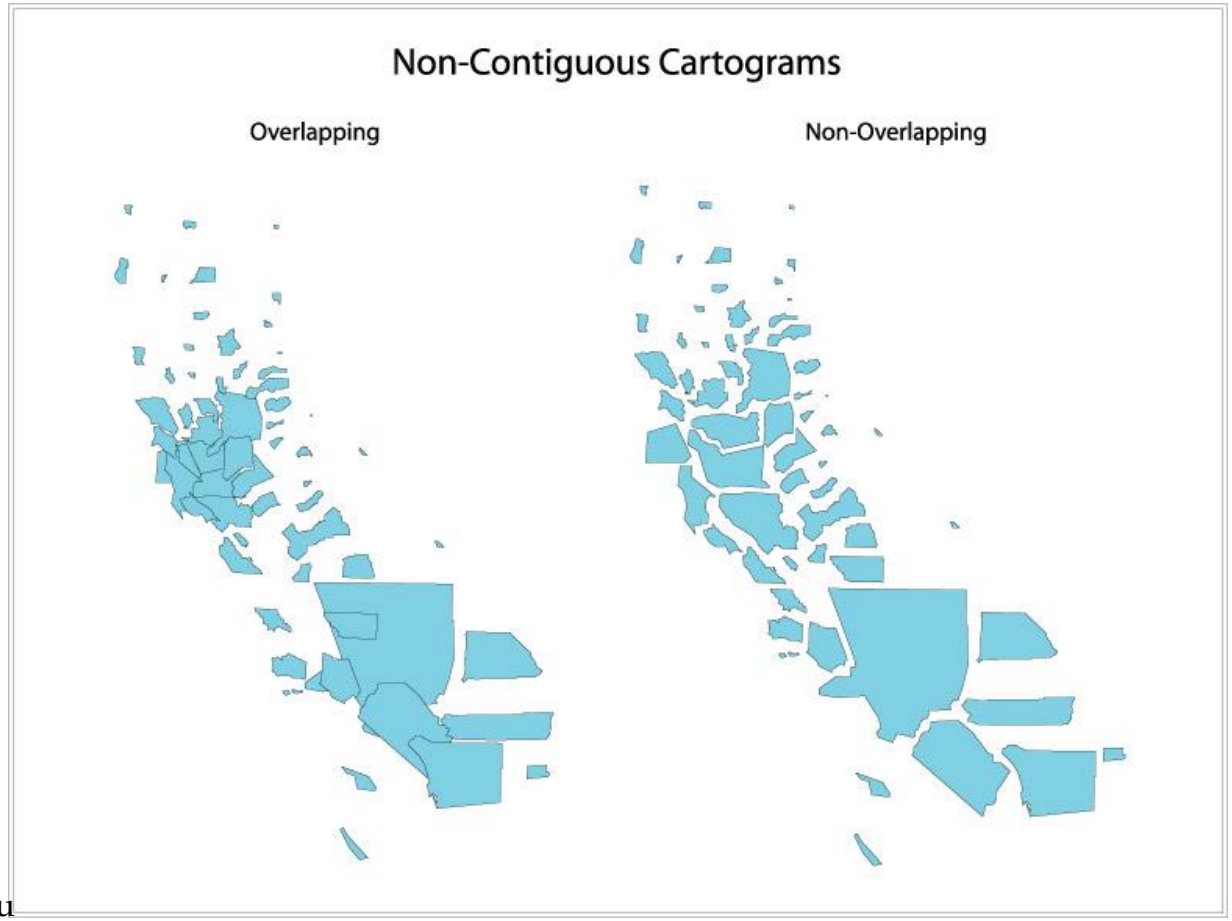
Contiguous

- County-level population



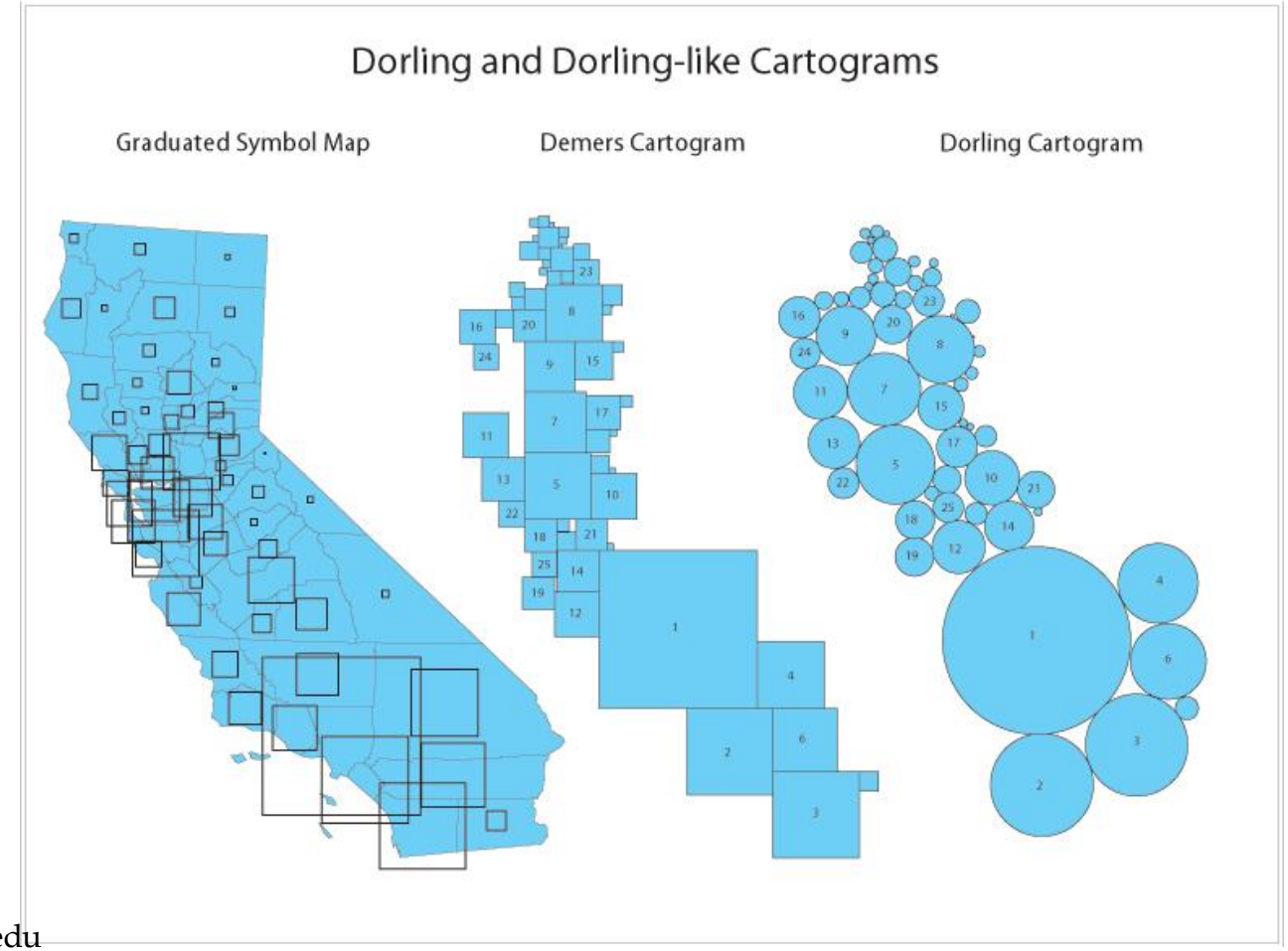
Source: ncgia.ucsb.edu

Non-contiguous



Source: ncgia.ucsb.edu

Dorling and Dorling-like



Source: ncgia.ucsb.edu

Making effective maps

Choosing a map type (symbols, choropleth, isarithmic, cartogram, …)

- Understand **where** and **how** phenomena changes (discrete/continuous, abrupt/smooth)
- Be wary of limitations of choropleths

Choose an appropriate data model

- **Normalize!** By age, wealth, population, etc.
- Understand your data and choose appropriate classification/binning method
  - Equal interval, quantile, mean-stdev, Jenk's.

- Understand the trade-offs of the projection you use

Assignment 2

Choose a dataset. A default option has been provided, but students may choose data of interest.

Make a lightweight geospatial dashboard, using Svetle + D3 to implement interactive selections and crossfiltering.

One view in the dashboard should be a choropleth map with color scale and map projection chosen carefully (you should be able to justify based on what we've learned).

Students may re-use and extend the code from our in-class demonstration.